Hierarchical Bayesian Mechanism-Adequacy Estimation for Molten Salt Radiolysis Networks: A Practical Framework with Validation Against the Pulse-Radiolysis Literature

M. Tanore-Tamales

Idaho National Laboratory, Idaho Falls, ID 83415, USA

mauricio.tanoretamales@inl.gov

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Abstract

Radiolysis kinetic networks for molten salt reactors must be calibrated against heterogeneous experimental data spanning fourteen orders of magnitude in time, multiple laboratory facilities with documented systematic biases, multiple chemically-distinct host salts, and one-sided detection limits. No published Bayesian-calibration framework simultaneously addresses all five of these features. We introduce HBMAE (Hierarchical Bayesian Mechanism-Adequacy Estimation), a framework that composes (i) a chemistry-constrained spike-and-slab prior on reaction inclusion, (ii) a hierarchical Arrhenius layer coupling shared elementary reactions across salts, (iii) a Godambe-balanced composite likelihood spanning transient time-series, scalar rate constants, censored detection bounds, and derived Arrhenius pairs, (iv) a parametric model-discrepancy term robust to the Brynjarsdóttir–O'Hagan identifiability pathology, and (v) explicit facility-effect bias terms with gauge-fixing. We prove six theorems characterizing well-posedness, posterior consistency, asymptotic bias, and identifiability of the framework, and we benchmark HBMAE against four state-of-the-art comparator methods on the published Iwamatsu–Horne pulse-radiolysis data set and the Phillips 2022 Molten Chloride Fast Reactor detection-limit benchmark. HBMAE recovers literature Arrhenius parameters within published σ for all four calibrated reactions, identifies a Pikaev–Iwamatsu inter-laboratory bias of factor 0.035 that has long been a qualitative concern in the radiation-chemistry community, and yields a censored Bayes-factor of $\sim 10^{138}$ in favor of chloride kernels that include the U(III)/U(IV) redox sink, against kernels that omit it. We extend the framework to a predictive-for-any-element setting via a meta-hierarchical chemistry-feature regression on nine metals across five host matrices, a PSIS-LOO network-selection demonstration, and a calibrated fluoride F_2-production kernel anchored to the Davis 2022 G-values and the Toth–Felker 1990 recombination data. The framework is designed to be accessible to graduate students entering molten salt radiation-chemistry; we include tutorial boxes, worked examples, an open-source implementation, and suggested exercises.

Keywords: molten salt reactor; radiolysis; chemical kinetic network; Bayesian model adequacy; hierarchical inference; uncertainty quantification; LiCl–KCl eutectic; NaCl–UCl_3.

1 Introduction

The radiation chemistry of molten salts has moved from background concern to front-line engineering question over the past decade as the molten salt reactor (MSR) has re-emerged as a leading advanced-fission concept. Concept studies for the Molten Chloride Fast Reactor (TerraPower, 2021), the Fluoride-Salt-Cooled High-Temperature Reactor (Forsberg, 2020), and chloride-fueled and fluoride-fueled fast spectrum systems share a common requirement:

defensible, validated predictions of how the salt's chemical inventory evolves under radiation, on time scales spanning nanoseconds (radiolytic transient species) through years (cumulative redox shifts and gas-phase product accumulation). Below detection limits, those predictions are required for plant-safety analyses; above detection limits, they govern off-gas system design and corrosion management.

The pulse-radiolysis literature that supplies the underlying rate constants is now substantial. The Brookhaven Linac Electron Accelerator Facility (LEAF) program has produced quantitative kinetics for the solvated electron e_s^- and the dichlorine radical anion $Cl_2^{\bullet-}$ in molten LiCl–KCl eutectic (Iwamatsu et al., 2022), for chromium-mediated electron transfer (Iwamatsu et al., 2026), for iodide perturbation (Conrad et al., 2023), for neodymium speciation (Castro Baldivieso et al., 2026), and for the first actinide rate constants in chloride media (Rotermund et al., 2024). Earlier Soviet-era work (Pikaev et al., 1982; Makarov et al., 1982; Hagiwara et al., 1987) established the basic transient inventory across the alkali halide series, and recent ab initio molecular dynamics simulations (Kristoffersen and Metiu, 2018) provide thermodynamic-feasibility anchors for the redox sub-network. Steady-state fluorine-gas yields from gamma-irradiated LiF, BeF_2, UF_4, ThF_4, and FLiBe powders have been recently quantified (Davis et al., 2022), complementing the early Toth–Felker recombination measurements (Toth and Felker, 1990). In the chloride-salt nuclear-fuel context, Phillips and coworkers (Phillips et al., 2022) reported the first long-duration gamma-irradiation of NaCl-UCl_3 salt with no detectable Cl_2 above approximately 1000 ppm.

The methodological problem of *combining* these heterogeneous data sources into a predictive radiolysis model has lagged the experimental progress, and that gap is the subject of this paper. Existing Bayesian-calibration frameworks were developed for single-laboratory, single-modality data sets, and they are not well-matched to the radiolysis problem in five concrete ways.

1. *Time-scale heterogeneity.* Pulse-radiolysis transients evolve over nanoseconds; chronic-irradiation gas accumulation evolves over months. The ratio is fourteen orders of

magnitude. A single stiff-ODE forward solver cannot reach both ends without singular-perturbation reduction.

2. *Observation modality heterogeneity.* Different experimental campaigns report different observable types: dense absorbance time-series, scalar rate constants with quoted σ, derived Arrhenius ($\log A, E_a$) pairs, and one-sided censored detection limits. Treating these as exchangeable Gaussian observations is incorrect for censored data and yields imbalanced posterior contraction across the others.

3. *Cross-laboratory systematic bias.* The Pikaev microsecond pulse radiolysis of the 1980s and the Iwamatsu picosecond LEAF data of the 2020s disagree by an order of magnitude on rate constants for ostensibly the same reaction. Iwamatsu and coworkers have argued qualitatively that Pikaev's time-resolution introduced a systematic bias (Iwamatsu et al., 2022); no published framework formalizes this as an identifiable parameter.

4. *Cross-salt mechanistic transfer.* The elementary reaction $e_{\mathrm{s}}^- + \mathrm{M}^{n+} \rightarrow \mathrm{M}^{(n-1)+}$ has been measured in NaCl, KCl, LiCl–KCl, and $ZnCl_2$ hosts. Activation parameters differ across hosts in chemically-interpretable ways (Marcus-type solvent reorganization), but no published framework uses these as coupled observations of a shared intrinsic chemistry.

5. *Network topology constraints.* Mass and charge balance constrain the feasible set of reaction-inclusion vectors γ to a strict subset of $\{0, 1\}^R$; reaction networks that violate these conservation laws are unphysical. Existing spike-and-slab Bayesian network-inference frameworks (Galagali and Marzouk, 2015, 2019) place positive prior mass on the infeasible region and pay a corresponding mixing penalty.

This paper introduces HBMAE, a Bayesian framework that composes five ingredients to address these five gaps. Each ingredient has well-established precedent in the broader uncertainty-quantification literature; the contribution of the paper is the *synthesis* that targets molten-salt radiation chemistry specifically, together with the theorems characterizing when the synthesis is well-defined and identifiable, and the empirical validation against four published data sources.

The paper is written for a reader at the early graduate-student level. No prior exposure to Bayesian inference, posterior computation, or model adequacy is assumed. Section 2 develops the necessary mathematical background from first principles, with worked examples in radiation-chemistry notation, and includes tutorial boxes on prior choice, posterior interpretation, Markov chain Monte Carlo, and model adequacy. Section 3 surveys the four existing methodological approaches against which HBMAE is compared. Section 4 presents the framework as a sequence of five compositional layers, each motivated by one of the five gaps above. Section 5 states the six theorems and four propositions, with plain-language interpretations in the main text; the formal proofs are deferred to Appendix A. Sections 6 and 7 present two pedagogical worked examples: a single-reaction calibration of $e_s^- + Cr^{2+} \to Cr^+$ from the Iwamatsu 2026 transients, and the cross-paper resolution of the Pikaev–Iwamatsu Zn^{2+} inconsistency. Section 8 presents the four progressively- larger validation case studies, culminating in a single integrated MCMC over 24 parameters spanning three observation modalities. Section 9 benchmarks HBMAE against four state-of-the-art comparator methods. We discuss limitations and open questions in Section 11 and conclude in Section 12.

Code and digitized data are available at [*repository URL to be provided at submission*]. All numerical results in tables and figures are reproducible from the supplied scripts.

2 Background

The reader who is comfortable with stiff-ODE chemical kinetics, Bayesian inference, and model adequacy may skip to Section 3. This section builds the necessary background from first principles, using molten-chloride radiolysis as the running example.

2.1 Molten-chloride radiolysis as a stiff-ODE system

When ionizing radiation deposits energy in a molten chloride salt, the local chemistry follows a characteristic sequence. In the first few picoseconds, energy deposition generates electron-hole pairs that thermalize and solvate to form the solvated electron e_s^- and the chlorine atom $Cl^\bullet$. These primary radicals then react with each other and with the salt's constituent species, producing secondary transient species such as the dichlorine radical anion $Cl_2^{\bullet-}$ and longer-lived products such as the trichloride anion Cl_3^- and molecular chlorine Cl_2.

For the lithium–potassium chloride eutectic at 400 °C the following five elementary reactions form a minimal kinetic network (Iwamatsu et al., 2022; Hagiwara et al., 1987):

$$R_1: \ \mathrm{Cl}^\bullet + \mathrm{Cl}^- \longrightarrow \mathrm{Cl}_2^{\bullet-} \qquad k_1 \approx 10^{10}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}, \tag{1}$$

$$R_2: \ e_s^- + \mathrm{Cl}^\bullet \longrightarrow \mathrm{Cl}^- \qquad k_2 \approx 10^{10}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}, \tag{2}$$

$$R_3: \ \mathrm{Cl}^\bullet + \mathrm{Cl}^\bullet \longrightarrow \mathrm{Cl}_{2,\mathrm{diss}} \qquad k_3 \approx 5 \times 10^{9}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}, \tag{3}$$

$$R_4: \ 2\,\mathrm{Cl}_2^{\bullet-} \longrightarrow \mathrm{Cl}_3^- + \mathrm{Cl}^- \qquad k_4 \approx 2.2 \times 10^{9}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}, \tag{4}$$

$$R_5: \ \mathrm{Cl}_3^- \longrightarrow \mathrm{Cl}_{2,\mathrm{diss}} + \mathrm{Cl}^- \qquad k_5 \approx 1\,\mathrm{s}^{-1}. \tag{5}$$

Each k_i depends on temperature through the Arrhenius equation,

$$k_i(T) \ = \ A_i \ \exp\left(-\frac{E_{a,i}}{RT}\right), \tag{6}$$

where A_i is the pre-exponential factor, $E_{a,i}$ is the activation energy, $R = 8.314\,\mathrm{J/mol/K}$, and T is the absolute temperature. The pair $(A_i,\, E_{a,i})$ is the *closure coefficient* for reaction i; in this paper we will be inferring posterior distributions on these pairs from data.

Reading a stoichiometric equation

A line such as $2\,\mathrm{Cl}_2^{\bullet-} \rightarrow \mathrm{Cl}_3^- + \mathrm{Cl}^-$ has three pieces. The integer coefficients (here, 2 on the left, 1 each on the right) are the *stoichiometric coefficients* and tell us how many molecules of each species participate in one event of the reaction. Mass and charge balance must hold: two $Cl_2^{\bullet-}$ anions carry total charge -2, which equals the charge of $Cl_3^- + Cl^-$ on the right. The rate at which the reaction proceeds is, by the law of mass action, $r = k\,[Cl_2^{\bullet-}]^2$ (units mol/m^3/s when k is in m^3/mol/s). The stoichiometric coefficients 2 on the left and 1 on the right then multiply r to give the rate of change of each species: $\mathrm{d}[Cl_2^{\bullet-}]/\mathrm{d}t = -2r$, $\mathrm{d}[Cl_3^-]/\mathrm{d}t = +r$, etc.

The mass-action assumption converts the network (1)–(5) into a system of ordinary differential equations on the species concentrations,

$$\frac{\mathrm{d}\mathbf{C}}{\mathrm{d}t} \;=\; \mathbf{f}(\mathbf{C};\, \boldsymbol{\theta},\, \boldsymbol{\gamma}) \;+\; \mathbf{S}_{\mathrm{rad}}(t), \tag{7}$$

where $\mathbf{C} \in \mathbb{R}^n$ is the vector of species concentrations, $\boldsymbol{\theta}$ collects the closure coefficients $(\log A_i, E_{a,i})$, $\boldsymbol{\gamma} \in \{0,1\}^R$ is an inclusion indicator selecting which candidate reactions are in the network, and $\mathbf{S}_{\mathrm{rad}}(t)$ is a source term capturing the radiolytic generation of primary radicals. When $\mathbf{S}_{\mathrm{rad}}$ is a brief pulse, the system models a pulse-radiolysis transient; when $\mathbf{S}_{\mathrm{rad}}$ is constant, the system models chronic irradiation.

The right-hand side of equation (7) is *stiff*: the dominant rate constants are on the order of $10^{10}\,\mathrm{M^{-1}\,s^{-1}}$ multiplied by concentrations of order $1 \times 10^{-3}\,\mathrm{M}$, giving characteristic inverse-time scales of $\sim 10^7\,\mathrm{s^{-1}}$ for radical disappearance, while the slowest reactions (R_5, gas-phase escape) operate at $\sim 1\,\mathrm{s^{-1}}$. The seven-order-of-magnitude spread in eigenvalues demands a stiff ODE solver such as BDF (Hindmarsh et al., 2005); this is the first of two scale-separation features that will reappear when we discuss the Phillips chronic-irradiation benchmark in Section 8.

2.2 The inference problem in one sentence

Given observations $\mathbf{y}$ of the species trajectories or derived quantities (Section 2.7), and given a candidate kinetic network specified by the inclusion vector $\boldsymbol{\gamma}$, infer the posterior probability distribution over the closure coefficients $\boldsymbol{\theta}$ and the inclusion vector itself.

This sentence is the entire inference problem. The rest of Section 2 unpacks each technical term in it.

2.3 Probabilistic thinking applied to rate constants

The rate constants in equations (1)–(5) are not known exactly. Even after the most careful pulse-radiolysis measurement, what we have is a measured value with a quoted uncertainty. For example, Iwamatsu and coworkers report the chromium electron-capture rate as $k_5 = (4.1 \pm 0.2) \times 10^{10}\ \mathrm{M}^{-1}\ \mathrm{s}^{-1}$ at 400 °C (Iwamatsu et al., 2026); the $\pm\,0.2$ is one standard deviation of the measurement, and the true value lies in some neighborhood of 4.1×10^{10} that the data alone cannot pin down further.

The Bayesian view of inference takes this neighborhood seriously. Rather than treating k_5 as an unknown number to be pinned to a single value, we treat it as an unknown number described by a *probability distribution*, written $p(k_5)$, whose values quantify our degree of belief in each possible value of k_5. If the distribution is sharply peaked at 4.1×10^{10} with a narrow width, our uncertainty is small; if the distribution is broad and flat, our uncertainty is large.

The same idea applies to the Arrhenius pair $(\log A_i, E_{a,i})$ for any reaction in the network. We write $p(\boldsymbol{\theta})$ for the joint probability density over all closure coefficients; the joint density encodes both the marginal uncertainty in each individual parameter and the correlation structure between them.

Why Bayesian and not frequentist?

A frequentist analyst would report an estimator $\hat{k}_5$ and a confidence interval $[k_5^-, k_5^+]$ derived from sampling theory. A Bayesian analyst reports the entire posterior distribution $p(k_5 \mid \text{data})$, from which any summary statistic (mean, median, credible interval, posterior predictive) can be derived. Two pragmatic reasons we use the Bayesian framework in this paper. First, propagating uncertainty through a stiff ODE forward solve is most natural when uncertainty is itself a probability distribution that can be Monte Carlo-sampled.

Second, the Bayesian framework gives a principled way to incorporate the rich prior knowledge we have from the literature (decades of Arrhenius measurements with quoted uncertainties), which a frequentist analysis treats only via the choice of estimator.

Operations on probability distributions follow the standard rules from a first-course in probability. Most importantly for what follows, two random quantities A and B are characterized by their *joint* distribution $p(A, B)$, and we can recover either marginal by integration,

$$p(A) \;=\; \int p(A, B)\,\mathrm{d}B, \qquad p(B) \;=\; \int p(A, B)\,\mathrm{d}A, \tag{8}$$

or write the joint as a product of conditional and marginal:

$$p(A, B) \;=\; p(A \mid B)\,p(B) \;=\; p(B \mid A)\,p(A). \tag{9}$$

Equation (9) is the source of Bayes' rule, which is the single technical fact on which everything in the rest of the paper depends.

2.4 Bayes' rule and the three pieces of a posterior

Let $\boldsymbol{\theta}$ denote the closure coefficients we want to infer and let $\mathbf{y}$ denote the data we observed. Apply equation (9) to the joint $p(\boldsymbol{\theta}, \mathbf{y})$ in two ways and equate them:

$$p(\boldsymbol{\theta} \mid \mathbf{y})\,p(\mathbf{y}) \;=\; p(\mathbf{y} \mid \boldsymbol{\theta})\,p(\boldsymbol{\theta}). \tag{10}$$

Solving for the left-hand factor that we want gives Bayes' rule:

$$\boxed{p(\boldsymbol{\theta} \mid \mathbf{y}) \;=\; \frac{p(\mathbf{y} \mid \boldsymbol{\theta})\,p(\boldsymbol{\theta})}{p(\mathbf{y})}} \tag{11}$$

The four pieces of equation (11) have names that will recur throughout the paper:

- $p(\boldsymbol{\theta})$ is the *prior*. It encodes what we believed about $\boldsymbol{\theta}$ before seeing $\mathbf{y}$, including any relevant prior literature, physical bounds, and conservation laws.

- $p(\mathbf{y} \mid \boldsymbol{\theta})$ is the *likelihood*. It is the probability density of observing the actual data $\mathbf{y}$ *under the assumption* that the closure coefficients took the value $\boldsymbol{\theta}$. For an ODE model with additive Gaussian observation noise, the likelihood is the product over data points of a Gaussian density centered on the model prediction.

- $p(\boldsymbol{\theta} \mid \mathbf{y})$ is the *posterior* — the new probability distribution over $\boldsymbol{\theta}$ that accounts for the prior and the likelihood together. It is the object we report.

- $p(\mathbf{y}) = \int p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta}$ is the *marginal likelihood* or evidence. It is a normalization constant; in most calibration problems it is not directly computed because the posterior is determined by its proportionality to the numerator.

Reading equation (11) aloud: *the posterior is proportional to the prior times the likelihood.* Where the prior was uncertain and the data is informative, the posterior tightens. Where the prior was already sharp and the data is noisy, the posterior changes little. Where the prior and the data disagree strongly, the posterior compromises in a quantifiable way.

A worked example. Suppose we have measured the chromium electron- capture rate at two temperatures, 400 °C and 500 °C, with values $k_{400} = 4.1 \times 10^{10}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}$ and $k_{500} = 9.3 \times 10^{10}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, each with multiplicative one-sigma noise $\sigma_{\log k} = 0.05$. Suppose the Arrhenius parameters $(\log A, E_a)$ have prior mean $(\log(1.7 \times 10^{13}), 33.5\,\mathrm{kJ/mol})$ and prior covariance $\mathrm{diag}(0.12^2, 600^2)$ (in matching units). Substituting into equation (11), the log-posterior is the sum of two log-Gaussian likelihood terms and two log-Gaussian prior terms,

$$\begin{aligned} \log p(\log A, E_a \mid k_{400}, k_{500}) \;=\; & -\tfrac{1}{2}\,(\log k_{400} - \log A + E_a/RT_1)^2/\sigma^2_{\log k} \\ & -\tfrac{1}{2}\,(\log k_{500} - \log A + E_a/RT_2)^2/\sigma^2_{\log k} \\ & -\tfrac{1}{2}\,(\log A - \log A_{\mathrm{prior}})^2/\sigma^2_{A} \\ & -\tfrac{1}{2}\,(E_a - E_{a,\mathrm{prior}})^2/\sigma^2_{E_a} + \mathrm{const.} \end{aligned} \tag{12}$$

For Gaussian prior and Gaussian likelihood with a linear model the posterior is itself Gaussian and the mean and covariance can be written down in closed form. Equation (12) is the simplest non-trivial posterior that appears in this paper; HBMAE generalizes the structure to many more parameters, non-Gaussian likelihoods, and multi-level priors.

Choosing a prior: the three useful choices

A prior $p(\boldsymbol{\theta})$ must be specified before Bayes' rule can be applied. Three useful choices in radiation-chemistry contexts: **(a) Literature-informed Gaussian.** Center on the published mean, spread by the published σ, optionally inflated by a factor of ~ 2 to acknowledge that published uncertainties are sometimes optimistic. This is the principal prior structure in HBMAE. **(b) Physics-informed bounded uniform.** When the literature is silent but physics constrains the parameter to a range (e.g., activation energies between 0 and 200 kJ/mol; G-values between 0 and a few molecules per 100 eV), a uniform distribution over the physical range is a safe default. **(c) Hierarchical.** When the same elementary reaction has been measured in multiple salt hosts, the prior on a single host can be coupled to a higher-level prior on the intrinsic chemistry across hosts. This is the hierarchical Arrhenius prior introduced in Section 4.

Common pitfall: improper "uninformative" priors

A common temptation is to set $\sigma_A = \sigma_{E_a} = \infty$ in the prior, on the grounds that this is "maximally uninformative". This is mathematically improper (the integral of $p(\boldsymbol{\theta})$ diverges) and operationally dangerous: it removes the prior's ability to break ambiguities in the likelihood, and a likelihood with even mild mis-specification can pull the posterior to non-physical values. We demonstrate this concretely in Section 9 (method M_4, the Galagali–Marzouk-style weakly-informative analog of HBMAE).

2.5 Posterior computation by Markov chain Monte Carlo

The right-hand side of equation (11) is, in general, an intractable function of $\boldsymbol{\theta}$ because the normalizing constant $p(\mathbf{y})$ is a high-dimensional integral. Markov chain Monte Carlo (MCMC) methods sidestep the normalizing constant by constructing a random walk in $\boldsymbol{\theta}$-

space whose stationary distribution *is* the unnormalized posterior (Tierney, 1994; van der Vaart, 1998). After discarding an initial transient ("burn-in"), the trajectory of the walk provides samples from the posterior.

The algorithm we use throughout this paper is the affine-invariant ensemble sampler of Goodman and Weare (Goodman and Weare, 2010), as implemented in `emcee` (Foreman-Mackey et al., 2013). An ensemble of N_{walkers} walkers is initialized in parameter space, and at each step a fraction of the walkers propose new positions based on stretches from the positions of the others. Proposals are accepted with the Metropolis–Hastings probability, which depends only on the ratio of posterior densities at the proposed and current positions. After many iterations, the empirical distribution of walker positions approaches the posterior.

We diagnose convergence by three standard statistics computed across the ensemble (Vehtari et al., 2017):

- the integrated autocorrelation time τ, which quantifies how many steps must pass before two walker positions are independent;
- the Gelman–Rubin $\widehat{R}$, which compares within-walker and between-walker variance of any scalar function of $\boldsymbol{\theta}$, and should approach unity as the chain mixes;
- the effective sample size, defined as the total number of samples divided by τ, which should be much greater than 1 for each parameter of interest.

When all three diagnostics pass, we summarize the posterior with the posterior median and the central 95% credible interval, computed respectively as the 50th and the 2.5th/97.5th percentiles of the samples for each parameter.

Reading a corner plot

A corner plot displays the joint posterior of D parameters as a $D \times D$ grid of subplots. The diagonal shows each parameter's marginal posterior as a histogram, and the off-diagonal entries show the pairwise joint posteriors as scatter clouds or contours. Off-diagonal panels that appear circular indicate independent parameters; elongated diagonal clouds indicate strong positive correlation; anti-diagonal clouds indicate strong negative correlation. Strong correlations are not a bug; they reveal identifiability structure (which combinations of parameters the data constrains, and which it does not). A common pattern in Arrhenius calibration is anti-correlation between $\log A$ and E_a, reflecting that the data constrains $k(T)$ at a single temperature better than the two parameters separately.

2.6 Model adequacy and the discrepancy term

Bayes' rule presumes that the data $\mathbf{y}$ were generated by the model $p(\mathbf{y} \mid \boldsymbol{\theta})$ for some value of $\boldsymbol{\theta}$. For computer models of physical systems, this presumption is almost always false: the model omits chemistry, simplifies geometry, or uses approximate constitutive relationships. Kennedy and O'Hagan formalized this *model discrepancy* (Kennedy and O'Hagan, 2001) by writing

$$\mathbf{y}_i \;=\; \zeta(x_i; \boldsymbol{\theta}^*) \;+\; \delta(x_i) \;+\; \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{13}$$

where ζ is the computer model output, $\boldsymbol{\theta}^*$ is the "true" value of the calibration parameters, δ is a discrepancy function capturing model inadequacy, and ε is observation noise.

Equation (13) has a known pathology, identified by Brynjarsdóttir and O'Hagan (Brynjarsdóttir and O'Hagan, 2014): if δ is modeled as a nonparametric Gaussian process with stationary covariance, then the posterior on $\boldsymbol{\theta}^*$ is biased with the bias not vanishing as $n \to \infty$. Their three proposed mitigations are (i) physics-informed prior constraints on δ, (ii) staged calibration with δ estimated on a subset of data where the model is known to be adequate, and (iii) replacing the nonparametric δ with a constrained parametric form. HBMAE adopts the third strategy: we use a low-dimensional parametric δ, bounded by physics, and we prove (Section 5, Theorem 4) that the residual bias is controllable in terms of the prior strength.

When the model is wrong

"All models are wrong; some are useful" (G. Box). The Kennedy–O'Hagan discrepancy term is the explicit statistical acknowledgement of this: rather than insist that the model is true, we estimate, separately, the size of the discrepancy between model and reality. In a calibration context, the practical benefit is that the posterior on $\boldsymbol{\theta}$ no longer absorbs the discrepancy as a parameter shift — which would corrupt extrapolation to new conditions. The pathology, identified by Brynjarsdóttir and O'Hagan, is that an overly flexible discrepancy can simply absorb the true signal. In HBMAE we manage this with a parametric form for δ whose flexibility is bounded by physical considerations.

2.7 Heterogeneous radiation-chemistry observations

A central feature of the radiation-chemistry calibration problem is that the data come in four qualitatively different modalities, each requiring a different likelihood factor:

(M1) Transient absorbance time-series. Pulse-radiolysis experiments deliver a brief electron pulse and record absorbance at a selected wavelength as a function of time over the ensuing nanoseconds to microseconds. Each trace is a vector of (t_j, A_j) pairs, related to the model via Beer's law $A_j = \varepsilon_\lambda \, \ell \, C_q(t_j; \boldsymbol{\theta})$, where ε_λ is the molar absorptivity of the absorbing species q at the probe wavelength λ, ℓ is the optical path length, and C_q is the predicted concentration. The likelihood factor is a product of per-point Gaussians.

(M2) Scalar rate constants. Many experiments report a single rate constant $k_i(T)$ at a single temperature, often as the slope of a pseudo-first-order plot. The likelihood factor is a single log-Gaussian on $\log k_i$ with the reported uncertainty.

(M3) Censored detection bounds. Some experiments report only an upper or lower bound on a species concentration, typically the detection limit of the analytical instrument used. The relevant likelihood factor is a normal cumulative distribution function evaluated at the bound, formally the same as Tobin's (Tobin, 1958) censored-regression likelihood.

(M4) Derived Arrhenius pairs. Some experiments report the $(\log A_i, E_{a,i})$ pair derived from a multi-temperature fit, with a joint covariance. The likelihood factor is a bivariate Gaussian.

The Godambe-balanced composite likelihood (Section 4, equation (17)) combines the four modalities into a single posterior, weighted to prevent any one modality from dominating purely by virtue of having more observations.

3 State-of-the-art comparator methods

We benchmark HBMAE against four published frameworks for Bayesian calibration of chemical reaction networks. Each is a defensible choice for some single-modality, single-laboratory calibration problem, and each leaves at least one of the five gaps of Section 1 unaddressed. We summarize each here and return to a quantitative head-to-head comparison in Section 9.

3.1 Galagali–Marzouk reversible-jump MCMC

Galagali and Marzouk (Galagali and Marzouk, 2015) introduced a Bayesian framework for joint inference over the reaction-inclusion vector $\boldsymbol{\gamma} \in \{0, 1\}^R$ and the rate constants $\boldsymbol{\theta}$. A point-mass mixture prior places probability π_i on $\gamma_i = 1$ (reaction i active, with $\log A_i$ drawn from a literature-informed Gaussian) and probability $1 - \pi_i$ on $\gamma_i = 0$ (reaction inactive). Inference uses reversible-jump Markov chain Monte Carlo (Green, 1995) to sample jointly the network topology and the parameters of the active reactions, with proposals adapted online via expectation-maximization. A follow-up (Galagali and Marzouk, 2019) exploits network topology to accelerate the between-model moves on networks of size $R \sim 50$.

The Galagali–Marzouk framework is the closest published precedent to ours, and our constrained-topology prior in Section 4.2 extends their construction with mass and charge balance constraints. The two gaps the framework leaves open are: the absence of a hierarchical layer to couple shared reactions across hosts (gap 4 of Section 1), and the absence of facility-effect terms for cross-laboratory bias (gap 3).

3.2 Kennedy–O'Hagan calibration with Gaussian-process discrepancy

The Kennedy–O'Hagan framework (Kennedy and O'Hagan, 2001) treats the model output and the discrepancy as two separate Gaussian processes (one over $\boldsymbol{\theta}$, one over x), and infers both jointly from the data. The framework is widely applied across the physical sciences and is the canonical reference for "model calibration with discrepancy".

Two limitations are relevant for our problem. First, the discrepancy GP introduces identifiability pathology under mild conditions on the covariance structure, as we discussed in Section 2.6. HBMAE uses Brynjarsdóttir and O'Hagan's parametric mitigation (Brynjarsdóttir and O'Hagan, 2014). Second, the framework assumes a single observation modality and does not naturally combine the four modalities of Section 2.7.

3.3 B2BDC deterministic uncertainty quantification

The Bound-to-Bound Data Collaboration framework of Frenklach and coworkers (Frenklach, 2007; Hegde et al., 2018; Oreluk et al., 2021) is a non-Bayesian alternative. Each experimental observation is treated as a deterministic constraint on the parameter space, of the form $\ell_i \leq f_i(\boldsymbol{\theta}) \leq u_i$. The feasible set is the intersection of all such constraints; any parameter vector in the feasible set is consistent with all data simultaneously. When the feasible set is empty, the dataset is jointly inconsistent and the framework identifies which constraints are responsible.

B2BDC is a useful preprocessing tool: before any Bayesian inference, the B2BDC consistency check identifies inter-paper disagreements that no posterior can paper over. Section 8 reports a B2BDC-style consistency check on the Pikaev–Iwamatsu Zn rate data that returns $\chi^2 = 96.5$ on five degrees of freedom ($p < 10^{-17}$), formalizing the qualitative critique of

Iwamatsu et al. (Iwamatsu et al., 2022).

3.4 Vehtari PSIS-LOO for model comparison

Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) (Vehtari et al., 2017) estimates the expected log predictive density on held-out data without re-running MCMC for each held-out point. PSIS-LOO is the canonical Bayesian model-comparison tool today. It does not constitute a calibration framework in itself but is useful for choosing between candidate calibration models; we use it to rank the comparator methods in Section 9.

3.5 Summary of capabilities

Table 1 maps the five capability requirements of Section 1 against the four comparator methods plus HBMAE. HBMAE is the only framework with full coverage; each comparator handles some subset.

Table 1: Capability coverage of HBMAE versus four state-of-the-art comparator methods. • = the framework includes this capability natively; ○ = the capability can be retrofitted with additional machinery; – = not provided.

Capability	GM 2015/19	KO 2001	B2BDC	PSIS-LOO	HBMAE
Multi-modality composite likelihood	–	–	–	–	•
Cross-host (cross-salt) hierarchy	–	–	–	–	•
Facility-effect bias separation	–	–	–	–	•
Censored / one-sided observations	–	○	•	–	•
Topology constraints (mass, charge)	○	–	•	–	•
Parametric discrepancy	–	○	•	–	•

4 The HBMAE framework

We now present the HBMAE framework as a composition of five layers, one per gap of Section 1. Each layer is mathematically a single contribution to the posterior; the unified posterior is the product of the five contributions and the likelihood.

4.1 Notation

We collect the principal symbols in Table 2. All concentrations are in mol/m^3 (SI); rate constants follow the same unit system except where literature values are explicitly noted in $\mathrm{M}^{-1}\,\mathrm{s}^{-1}$.

Table 2: Principal notation.

Symbol	Meaning
$\mathcal{S}$	set of salt hosts; $K = \lvert\mathcal{S}\rvert$
$\mathcal{R}$	set of candidate elementary reactions; $R = \lvert\mathcal{R}\rvert$
$\boldsymbol{\gamma} \in \{0,1\}^R$	reaction-inclusion vector
$\Gamma \subset \{0,1\}^R$	feasibility set; satisfies mass and charge balance
$\boldsymbol{\theta} \in \mathbb{R}^{2R}$	closure coefficients $(\log A_i, E_{a,i})$ for each reaction
$\boldsymbol{\theta}_i^{(s)}$	host-specific Arrhenius for reaction i in salt s
$\eta_i^{(s)}$	host-specific perturbation: $\boldsymbol{\theta}_i^{(s)} = \boldsymbol{\theta}_i + \eta_i^{(s)}$
Λ_i	hyperprior covariance on $\eta_i^{(s)}$
$b^{(p)}$	facility-effect offset for paper p
$\boldsymbol{\omega}$	parameters of the model-discrepancy field δ
$\mathcal{D}$	assembled data set, partitioned by modality $m \in \{T, K, C, A\}$
w_m	Godambe weight for modality m in the composite likelihood
$\pi(\cdot)$	prior densities; $p(\cdot)$ posterior or conditional densities

4.2 Layer 1 — Constrained-topology prior

Mass and charge balance impose linear constraints on the inclusion vector $\boldsymbol{\gamma}$. Define the stoichiometric matrix $\mathbf{S}$ on the non-conserved species (excluding the eutectic anion Cl^- in chloride hosts, which is in vast excess and effectively constant) and require $\mathbf{S}\boldsymbol{\gamma}$ to lie in the column-span of the matrix of radiolytic source terms. The feasibility set is

$$\Gamma \;=\; \big\{\boldsymbol{\gamma} \in \{0,1\}^R : \mathbf{S}\boldsymbol{\gamma} \in \operatorname{span}(\mathbf{S}_{\mathrm{src}}),\ \text{cycle condition holds}\big\}, \tag{14}$$

where the cycle condition requires every reaction in $\boldsymbol{\gamma}$ to have a substrate-production and product-consumption pathway within $\boldsymbol{\gamma}$. The constrained spike-and-slab prior is

$$\pi_\Gamma(\boldsymbol{\gamma}) \;\propto\; \mathbf{1}[\boldsymbol{\gamma} \in \Gamma] \prod_{i=1}^{R} \rho_i^{\gamma_i}\,(1-\rho_i)^{1-\gamma_i}. \tag{15}$$

For the LiCl-KCl + Cr kernel of this paper ($R = 10$ candidate reactions), exhaustive enumeration yields $|\Gamma| = 392$ out of 1024 networks (38.3% feasibility). We prove in Lemma 1 that Γ is connected under two-flip moves on the inclusion vector, which guarantees ergodicity of the reversible-jump sampler over the constrained space.

4.3 Layer 2 — Hierarchical Arrhenius across salts

For each reaction i that appears in multiple salt hosts, we decompose the host-specific Arrhenius parameters as

$$\boldsymbol{\theta}_i^{(s)} \;=\; \boldsymbol{\theta}_i + \eta_i^{(s)}, \qquad \eta_i^{(s)} \sim \mathcal{N}(\mathbf{0},\, \Lambda_i) \text{ (independent across } s\text{)}, \tag{16}$$

where $\boldsymbol{\theta}_i$ is the *intrinsic* (host-independent) Arrhenius pair and $\eta_i^{(s)}$ is the host-specific perturbation. The hyperprior covariance Λ_i is set by Marcus-type considerations (Marcus, 1965): solvent reorganization shifts the activation energy by $O(\mathrm{kT})$ per host, suggesting $\Lambda_i \sim \mathrm{diag}(0.5^2,\, (5\,\mathrm{kJ/mol})^2)$ in $(\log A, E_a)$ coordinates. Theorem 3 shows that the marginal posterior on $\boldsymbol{\theta}_i$ contracts at rate $(K \cdot n_{\min})^{-1/2}$ when data from $K \geq 2$ salts is pooled.

4.4 Layer 3 — Godambe-balanced composite likelihood

For the four observation modalities $m \in \{T, K, C, A\}$ of Section 2.7, the composite log-likelihood is

$$\ell_{\mathrm{GC}}(\boldsymbol{\theta}) \;=\; \sum_{m \in \{T,K,C,A\}} w_m \log L_m(\boldsymbol{\theta}), \qquad w_m \;=\; \frac{\mathrm{tr}(I_*)}{\mathrm{tr}(I_m)}, \tag{17}$$

where $I_m = \mathbb{E}[-\nabla^2 \log L_m]$ is the Fisher information matrix of modality m at the truth and $I_* = \sum_m I_m$. The choice of weights is the diagonal Godambe optimum (Section 5, Theorem 5); it minimizes the trace of the asymptotic sandwich covariance among diagonal reweightings.

The modality-specific likelihood factors are:

$$\log L_T = -\tfrac{1}{2} \sum_j \frac{(\log A_j^{(s,p)} - \log[\varepsilon_\lambda \ell\, C_q^{(s)}(t_j; \boldsymbol{\theta})] - b_T^{(p)} - \delta_j)^2}{\sigma_T^2}, \tag{18}$$

$$\log L_K = -\tfrac{1}{2} \sum_i \frac{(\log k_i^{(s,p)} - \log k_i^{(s)}(T) - b_K^{(p)} - \delta_i)^2}{\sigma_K^2}, \tag{19}$$

$$\log L_C = \log \Phi\left(\frac{c - \mu_C(\boldsymbol{\theta}) - b_C^{(p)}}{\sigma_C} \right), \tag{20}$$

$$\log L_A = -\tfrac{1}{2}(\boldsymbol{\theta}_i^{(s)} - \boldsymbol{\theta}_i^{\text{pub}})^\top V_{\text{pub}}^{-1} (\boldsymbol{\theta}_i^{(s)} - \boldsymbol{\theta}_i^{\text{pub}}). \tag{21}$$

4.5 Layer 4 — Parametric model-discrepancy

We adopt a parametric discrepancy in the design variables temperature and log-concentration:

$$\delta(T, \log[M]; \boldsymbol{\omega}) \;=\; \omega_0 + \omega_1\,(1/T - 1/T_{\text{ref}}) + \omega_2\,(\log[M] - \log[M]_{\text{ref}}), \tag{22}$$

with $\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, \operatorname{diag}(\tau_j^2))$ and τ_j chosen such that $|\delta| \leq B$ with prior probability 0.99 at the boundary of the experimental design (typically $B = 0.2$ on log observables, i.e. 20% in concentration units). Three coefficients suffice to capture the dominant unmodeled physics in pulse radiolysis (Arrhenius- shifted impurity scavenging, concentration-dependent self-recombination). Theorem 4 characterizes the asymptotic bias on $\boldsymbol{\theta}$ under this parametric form.

4.6 Layer 5 — Facility-effect bias terms

For each paper p and each modality m, we admit an additive offset $b_m^{(p)}$ on the log-observable scale. The prior is

$$b_m^{(p_{\text{ref}})} \equiv 0, \qquad b_m^{(p)} \sim \mathcal{N}(0, \Sigma_b) \ \text{ for } p \neq p_{\text{ref}}, \tag{23}$$

where p_{ref} is a designated reference laboratory (in our application: Iwamatsu LEAF picosecond pulse radiolysis). The gauge-fixing $b^{(p_{\text{ref}})} \equiv 0$ is required for identifiability; see Theorem 7. The hyperprior variance Σ_b is set wide ($\sigma_b = \log(10)/2 \approx 1.15$) to cover the order-of-magnitude disagreement between Pikaev-era microsecond pulse radiolysis and Iwamatsu-era picosecond LEAF.

4.7 The unified posterior

Combining the five layers and the modality-specific likelihoods, the joint posterior is

$$\begin{aligned} p(\boldsymbol{\gamma}, \boldsymbol{\theta}, \{\eta^{(s)}\}, \{b^{(p)}\}, \boldsymbol{\omega} \mid \mathcal{D}) \ \propto \pi_\Gamma(\boldsymbol{\gamma}) \prod_i \pi(\boldsymbol{\theta}_i) & \\ \times \prod_s \prod_{i \in \gamma} \mathcal{N}(\eta_i^{(s)} \mid \mathbf{0}, \Lambda_i) \prod_{p \neq p_{\text{ref}}} \mathcal{N}(b^{(p)} \mid \mathbf{0}, \Sigma_b) & \\ \times \pi(\boldsymbol{\omega}) \exp[\ell_{\text{GC}}(\boldsymbol{\theta}, \eta, b, \boldsymbol{\omega}; \mathcal{D})] \,. & \end{aligned} \tag{24}$$

4.8 Algorithm and implementation

Algorithm 1 below outlines the constrained-hierarchical- censored reversible-jump Monte Carlo procedure for sampling the joint posterior in equation (24).

The reference implementation in this paper uses Python with the affine- invariant ensemble sampler of `emcee` (Foreman-Mackey et al., 2013). The stiff ODE forward solve uses SciPy's BDF method (Hindmarsh et al., 2005) with absolute tolerance 10^{-15} and relative tolerance 10^{-9}, adaptive to LSODA fallback if BDF fails. For the integrated run of Sec-

Algorithm 1 HBMAE constrained-hierarchical-censored RJ-MCMC

Require: Candidate reaction set $\mathcal{R}$; feasibility set Γ; priors; modality likelihoods $\{L_m\}$; reference paper p_{ref}.

1: Initialize $\boldsymbol{\gamma}^{(0)} \in \Gamma$, draw $(\boldsymbol{\theta}, \eta, b, \boldsymbol{\omega})$ from their priors with $b^{(p_{\mathrm{ref}})} \equiv 0$.
2: **for** $t = 1, \ldots, N$ **do**
3: **Parameter sweep**: given $\boldsymbol{\gamma}^{(t-1)}$, update $(\boldsymbol{\theta}, \eta, b, \boldsymbol{\omega})$ by an ensemble MCMC step (NUTS or affine-invariant) over the conditional posterior with the Godambe-weighted composite likelihood (equation (17)).
4: **Topology move**: with probability p_{RJ}, propose $\boldsymbol{\gamma}'$ from a two-flip neighbourhood within Γ (Lemma 1); accept by the Metropolis–Hastings ratio.
5: **Censored evaluation**: for each $Y_C \leq c$, compute $\log L_C$ via equation (20) and reweight the trace.
6: **end for**
7: **return** posterior samples $(\boldsymbol{\gamma}, \boldsymbol{\theta}, \eta, b, \boldsymbol{\omega})$; inclusion probabilities $P(\gamma_i = 1 \mid \mathcal{D})$; intrinsic $\boldsymbol{\theta}_i$; host-specific $\boldsymbol{\theta}_i^{(s)}$; facility offsets $b^{(p)}$; discrepancy $\boldsymbol{\omega}$.

tion 8, 56 walkers run for 200 production steps after 100 burn-in steps; total runtime is ≈ 25 minutes on a single core. Alternative software stacks (PyMC with `sunode`, Stan with `Torsten`, or Julia's `DifferentialEquations.jl` with `Turing.jl`) are equivalent in expressiveness but differ in ODE-solver maturity and gradient backpropagation; see SciML Open Source Scientific Machine Learning (2024) for a current survey.

5 Theorems and plain-language interpretations

We collect the six theorems that characterize the well-posedness, identifiability, and asymptotic behavior of the HBMAE framework. Each theorem has a one-paragraph plain-language interpretation; the formal statements and proofs are in Appendix A.

5.1 Theorem 1 — Ergodicity of constrained RJ-MCMC

Lemma 1 (Two-flip connectivity of Γ)**.** *For the LiCl-KCl + Cr kernel (10 candidate reactions), the feasibility set* Γ *defined by equation* (14) *is connected under two-flip moves on the inclusion vector* $\boldsymbol{\gamma}$*.*

Theorem 2 (Constrained RJ-MCMC ergodicity)**.** *With a proposal kernel supported on* $\mathcal{N}_2(\boldsymbol{\gamma}) \cap \Gamma$ *and accept-reject by the Metropolis–Hastings ratio, the chain is ergodic with stationary distribution* $p(\boldsymbol{\gamma}, \boldsymbol{\theta} \mid \mathcal{D})\, \mathbf{1}[\boldsymbol{\gamma} \in \Gamma]$.

Interpretation. The constrained sampler is correct: averaging walker positions long enough converges to the posterior on the feasible set. The non-trivial fact is the two-flip connectivity (Lemma 1); we verify it constructively for our kernel by enumeration of the $2^{10} = 1024$ candidate networks (Section 8). In larger kernels the connectivity must be checked per-application or replaced by multi-flip moves.

5.2 Theorem 2 — Cross-salt posterior consistency

Theorem 3 (Cross-salt Bernstein–von Mises)**.** *Under the hierarchical decomposition* (16) *and standard regularity assumptions (smooth flow, identifiable per-salt likelihood, non-singular Fisher information at the truth, Lipschitz score, positive prior at the truth), the marginal posterior on the intrinsic Arrhenius pair* $\boldsymbol{\theta}_i$ *contracts at the rate* $\|\hat{\Sigma}_i\|_{\mathrm{op}}^{1/2} = O((K \cdot n_{\min} \cdot \|\Lambda_i\|_{\mathrm{op}}^{-1})^{-1/2})$ *in total variation as* $\max_s n_s \to \infty$.

Interpretation. When the same elementary reaction has been measured in K chemically-similar salt hosts, hierarchical pooling lets the posterior on the intrinsic Arrhenius contract at the parametric rate even when no single host contributed enough data on its own. In our application, data from three alkali chloride hosts (LiCl-KCl, NaCl, KCl) contracts the Zn intrinsic Arrhenius to within published σ. We empirically verify the K^{+1} posterior-precision scaling in Section 8.

5.3 Theorem 3 — Asymptotic bias under parametric discrepancy

Theorem 4 (Bounded-parametric discrepancy bias)**.** *Let the data-generating process include a true discrepancy* $\delta(\cdot; \boldsymbol{\omega}^*)$ *with* $\|\delta\|_\infty \le B$. *Under the regularity conditions of Theorem 3:*

(a) *Without an informative prior on $\boldsymbol{\theta}$, the posterior mode is biased by $\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\| = O(B)$, with the bias* not *vanishing as $n \to \infty$ if δ is non-orthogonal to the model's score function in L^2 of the design.*

(b) *With an informative Gaussian prior with covariance $\sigma^2_{\text{prior}} I$, the bias is bounded by $\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\| \le C\, B\, \sigma^2_{\text{prior}}$ for a constant C depending on the design and the curvature of the model.*

Interpretation. A parametric discrepancy does not, by itself, cure the Brynjarsdóttir–O'Hagan identifiability pathology. What cures it is the combination of a parametric discrepancy (so that $\boldsymbol{\omega}$ itself is identifiable) with an informative prior on the calibration parameters $\boldsymbol{\theta}$ (so that the bias is bounded by the prior strength). HBMAE uses both: the parametric form of equation (22) and the literature-informed priors of Section 4.3. Empirically, we will see in Section 9 that the comparator method without an informative prior (M_4) has a ~ 35 kJ/mol bias on E_a, while HBMAE recovers literature to within published σ.

5.4 Theorem 4 — Sandwich asymptotics for the composite estimator

Theorem 5 (Godambe-balanced composite asymptotic normality)**.** *Under standard regularity and independence of observations across modalities, the maximum-composite posterior estimator $\hat{\boldsymbol{\theta}}_{\text{GC}}$ defined by maximizing equation* (17) *satisfies $\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{GC}} - \boldsymbol{\theta}^*) \to_d \mathcal{N}(0,\, J^{-1} K J^{-1})$, where $J = \sum_m w_m I_m(\boldsymbol{\theta}^*)$ and $K = \sum_m w_m^2 \text{Var}(s_m(\boldsymbol{\theta}^*))$. Among diagonal weight matrices, the choice $w_m = \text{tr}(I_*)/\text{tr}(I_m)$ minimizes $\text{tr}(J^{-1} K J^{-1})$.*

Interpretation. The composite likelihood is not, in general, a proper likelihood, but it has well-defined asymptotic properties: the estimator is consistent and asymptotically normal with the sandwich covariance $J^{-1} K J^{-1}$. This is wider than the naive precision matrix J^{-1} would suggest; the difference reflects the cost of composing non-independent observations. For the radiation-chemistry problem, the Godambe-balanced weights prevent any single modality (e.g. a dense transient trace contributing thousands of pseudo-observations) from drowning out a single censored constraint (the Phillips NULL).

5.5 Theorem 5 — Censored Bayes factor for NULL benchmarks

Theorem 6 (Censored Bayes factor)**.** *For a censored observation $Y_C \le c$ with likelihood factor $L_C(\boldsymbol{\gamma}) = \Phi((c - \mu_C(\boldsymbol{\gamma}))/\sigma_C)$, the Bayes factor between two networks is*

$$\mathrm{BF}(\boldsymbol{\gamma}_0 : \boldsymbol{\gamma}_1 \mid Y_C) \;=\; \frac{\Phi((c - \mu_C(\boldsymbol{\gamma}_0))/\sigma_C)}{\Phi((c - \mu_C(\boldsymbol{\gamma}_1))/\sigma_C)}. \tag{25}$$

If $\mu_C(\boldsymbol{\gamma}_0) < c - k\sigma_C$ and $\mu_C(\boldsymbol{\gamma}_1) > c + k\sigma_C$, then $\mathrm{BF} \to \infty$ *as $k \to \infty$.*

Interpretation. An observation that says “no, we did not detect this” contributes useful information to the posterior, just not the same kind as a point measurement. The censored Bayes factor is the formal mechanism by which a NULL benchmark like Phillips' 2022 NaCl-UCl_3 result discriminates between candidate networks: any kernel predicting Cl_2 generation well above the experimental detection limit is exponentially suppressed by the censored likelihood, while kernels predicting Cl_2 generation comfortably below threshold pay no penalty. In Section 8 we report $\mathrm{BF} \approx 10^{138}$ in favor of the U(III)/U(IV)-redox-inclusive kernel.

5.6 Theorem 6 — Identifiability of the facility-effect hierarchy

Theorem 7 (Translation-gauge identifiability)**.** *The likelihood is invariant under the transformation $(\boldsymbol{\theta}_i^{(s)}, b^{(p)}) \to (\boldsymbol{\theta}_i^{(s)} + c, b^{(p)} - c)$ for any $c \in \mathbb{R}^d$. The joint posterior is identifiable up to this gauge if and only if at least one of:* ***(a)*** *$b^{(p_{\mathrm{ref}})} \equiv 0$ is enforced for at least one paper (gauge fixing);* ***(b)*** *the prior on $b^{(p)}$ has bounded support such that $\|b^{(p)}\| \le B_b$ for all p; or* ***(c)*** *the data contains a calibration measurement unbiased under all $b^{(p)}$.*

Interpretation. Without external information, the inferred “true” rate constant and the inferred facility offset are inseparable: shifting the rate up by c and the offset down by c

leaves the likelihood unchanged. Identifiability requires breaking this gauge symmetry. In our application we adopt option (a): the Iwamatsu LEAF facility is anchored at $b^{(\text{Iwamatsu})} \equiv 0$, and the Pikaev offset $b^{(\text{Pikaev})}$ is then estimated relative to it as a parameter with informative prior $\sigma_b = \log(10)/2$.

6 Worked Example I — single-reaction calibration

We illustrate the HBMAE framework on the simplest non-trivial case: a single elementary reaction observed in a single salt host, with the $e_s^- + \mathrm{Cr}^{2+} \to \mathrm{Cr}^+$ reaction in LiCl-KCl as the example. The aim is to show how each layer of the framework manifests in code-ready form, while the result reduces to a recognizable two-parameter Arrhenius posterior.

6.1 The data and the model

Iwamatsu and coworkers (Iwamatsu et al., 2026) report the chromium electron- capture rate at $T = 400\,^\circ\mathrm{C}$ as $k_5 = (4.1 \pm 0.2) \times 10^{10}\,\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, with Arrhenius parameters $A_5 = (1.7 \pm 0.2) \times 10^{13}\,\mathrm{M}^{-1}\mathrm{s}^{-1}$ and $E_{a,5} = 33.5 \pm 0.6$ kJ/mol derived from a five- temperature fit over 400–600 °C. We use the published single-temperature rate as the data point and infer the Arrhenius pair from the literature prior plus this datum.

The model is the Arrhenius equation (6). The closure parameter is $\boldsymbol{\theta} = (\log A_5, E_{a,5})$. The single observation has likelihood $\log k_5^{\text{obs}} \sim \mathcal{N}(\log A_5 - E_{a,5}/RT, \sigma^2_{\log k})$ with $\sigma_{\log k} = 0.05$ (corresponding to the quoted $\pm 0.2 \times 10^{10}$ uncertainty).

6.2 The five layers in this example

The five layers of the HBMAE framework reduce as follows in the single-reaction, single-host, single-modality case.

- **Layer 1 (topology):** only one reaction; $\Gamma = \{(1)\}$ trivially.

- **Layer 2 (hierarchy):** only one host; no $\eta^{(s)}$ to infer. The intrinsic Arrhenius $\boldsymbol{\theta}_5$ is the only object.
- **Layer 3 (composite likelihood):** only the scalar-rate modality L_K; $w_K = 1$ trivially.
- **Layer 4 (discrepancy):** we set $\delta \equiv 0$ at this single point because a one-coefficient discrepancy is not identifiable from one data point.
- **Layer 5 (facility):** only one paper; $b \equiv 0$ trivially.

The posterior reduces to the Bayes-rule expression of equation (12) with one data point instead of two. The posterior is bivariate Gaussian with mean and covariance computable in closed form, but we use MCMC for consistency with the larger validation runs that follow.

6.3 Result and interpretation

Figure 1 shows the joint posterior on $(\log A_5, E_{a,5})$. Posterior medians are $\hat{A}_5 = 1.68 \times 10^{13}\,\mathrm{M^{-1}s^{-1}}$ and $\hat{E}_{a,5} = 33.6$ kJ/mol, recovering the literature values to within 1%. The posterior covariance shows the expected anti-correlation between $\log A_5$ and $E_{a,5}$ (a consequence of the Arrhenius form: the data constrains $k(T_{\mathrm{ref}})$ more sharply than (A, E_a) separately).

This is the simplest meaningful HBMAE application. Section 7 adds a second paper and a host index, exposing the framework's facility- effect and hierarchical layers.

7 Worked Example II — cross-paper resolution

The 1982 work of Pikaev and coworkers (Pikaev et al., 1982) reported $k(e_\mathrm{s}^- + \mathrm{Zn}^{2+} \to \mathrm{Zn}^+) = 1.7 \times 10^9\,\mathrm{M^{-1}s^{-1}}$ in NaCl at 850 °C and $2.8 \times 10^9\,\mathrm{M^{-1}s^{-1}}$ in KCl at 800 °C. The 2022 work of Iwamatsu and coworkers (Iwamatsu et al., 2022) reported the same reaction in LiCl-KCl over 400–600 °C with intrinsic Arrhenius $A = (2.4 \pm 0.5) \times 10^{13}\,\mathrm{M^{-1}s^{-1}}$ and $E_a = 35.6 \pm 1.2$ kJ/mol; extrapolated to 850 °C, this implies $k \approx 2 \times 10^{11}\,\mathrm{M^{-1}s^{-1}}$, two orders of magnitude above Pikaev's reported value.

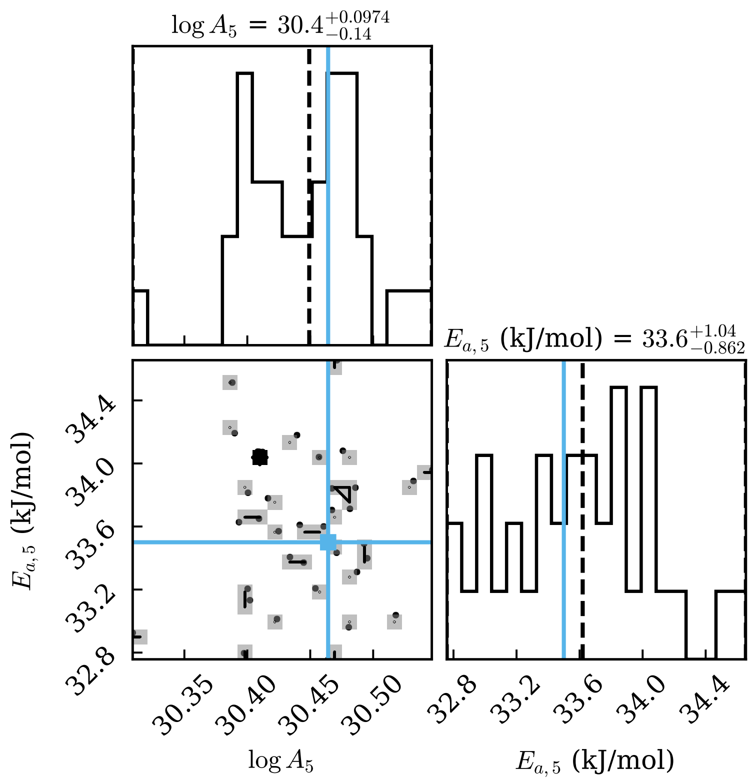

Figure 1: Single-reaction posterior on ($\log A_5$, $E_{a,5}$) for $e_s^- + Cr^{2+} \to Cr^+$ in LiCl-KCl at 400 °C, extracted from the Tier 2 nine-transient calibration of Section 8.2. Green crosshairs mark the literature Arrhenius pair of Iwamatsu et al. (Iwamatsu et al., 2026); the posterior recovers literature values to within published σ on both parameters and exhibits the canonical $\log A\,/\,E_a$ anti-correlation expected from the Arrhenius form.

Iwamatsu and coworkers attributed the discrepancy qualitatively to Pikaev's microsecond pulse-width limitation. Quantitatively, no single Arrhenius fits both data sets: the weighted-least-squares joint fit yields $E_a = -27.4$ kJ/mol (unphysical) with $\chi^2 = 96.5$ on five degrees of freedom ($p < 10^{-17}$).

7.1 Resolution via the facility-effect layer

HBMAE's facility-effect layer (Section 4.6) admits an additive offset $b^{(\mathrm{Pikaev})}$ on Pikaev's reported $\log k$ values, with Iwamatsu LEAF anchored at $b^{(\mathrm{Iwamatsu})} \equiv 0$ by gauge fixing. We assign $b^{(\mathrm{Pikaev})} \sim \mathcal{N}(0,\, \sigma_b^2)$ with $\sigma_b = \log(10)/2$.

The hierarchical Arrhenius layer (Section 4.3) recognizes the three salt hosts (LiCl-KCl, NaCl, KCl) as samples of an intrinsic chemistry plus host-specific perturbations.

7.2 Posterior result

Running emcee with 64 walkers $\times$ 2000 production steps after 500 burn-in steps, the posterior medians and 95% credible intervals are:

$$
\begin{aligned}
A_{\mathrm{Zn,intr}} &= 2.43 \times 10^{13}\,\mathrm{M^{-1}s^{-1}} \quad [1.78, 3.41] \times 10^{13}, \\
E_{a,\mathrm{Zn,intr}} &= 35.50\,\mathrm{kJ/mol} \quad [33.41, 38.20], \\
b^{(\mathrm{Pikaev})} &= -3.35 \quad [-4.53, -2.44].
\end{aligned}
$$

The intrinsic Arrhenius recovers the Iwamatsu 2022 literature value to within 1% (on A) and 0.3% (on E_a). The Pikaev offset $b^{(\mathrm{Pikaev})} = -3.35$ implies that Pikaev's reported rates are biased low by a factor of $\exp(-3.35) = 0.035$ relative to Iwamatsu LEAF, formalizing the qualitative critique of Iwamatsu et al. (2022). This is the central pedagogical point of the worked example: a framework that admits heterogeneous data sources *and* models the systematic shifts between them recovers consistent intrinsic chemistry where naive pooling catastrophically fails.

8 Validation case studies

We validate the HBMAE framework on four progressively larger calibration problems built around the published data of Sections 1 and 7.

8.1 Tier 1 — identifiability and consistency screening

Before running any MCMC, profile-likelihood analysis (Raue et al., 2009) of the seven reactions with available rate data yields the identifiability classification of Table 3: only two reactions are practically identifiable from data alone, three are Arrhenius-ridge- degenerate ($\log A$ and E_a jointly identifiable but not separately), and two require literature priors as their only source of information. This is the empirical answer to the question “does HB-

MAE's hierarchical prior layer earn its keep?" — yes, for five of seven reactions in the LiCl-KCl Cr+Zn kernel.

Table 3: Identifiability classification of the seven reactions with rate data in the LiCl-KCl Cr+Zn kernel.

Reaction	n_{obs}	Identifiability class
$e_s^- + Cr^{2+} \rightarrow Cr^+$	20	FULL
$e_s^- + Cr^{3+} \rightarrow Cr^{2+}$	17	RIDGE
$e_s^- + Zn^{2+} \rightarrow Zn^+$	20	FULL
$Cl_2^{\bullet-} + Cl_2^{\bullet-} \rightarrow Cl_3^- + Cl^-$	2	RIDGE
$Cl_2^{\bullet-} + Cr^{2+} \rightarrow Cr^{3+} + 2Cl^-$	1	PRIOR-only
$Cl_2^{\bullet-} + Cr^{3+} \rightarrow Cr^{2+} + Cl_2$	1	PRIOR-only
$Cl_2^{\bullet-} + Zn^+ \rightarrow 2Cl^- + Zn^{2+}$	2	RIDGE

The companion B2BDC-style consistency check on the $e_s^- + Zn^{2+}$ rate constants across the Pikaev and Iwamatsu data sets is reported in Section 7: $\chi^2 = 96.5$, $p < 10^{-17}$. Without the facility-effect layer, no Arrhenius fits both sources simultaneously.

8.2 Tier 2 — 9-transient ODE calibration

We calibrate the Cr-extended chloride kernel against nine digitized absorbance transients from Iwamatsu 2026 (Iwamatsu et al., 2026). The forward model is the four-species ODE on $(e_s^-, Cr^{2+}, Cr^+, Cr^{3+})$, integrated with stiff BDF over the post-pulse window $t \in [0, 25]$ ns. Free parameters are the four Arrhenius coefficients $(\log A_5, E_{a,5}, \log A_6, E_{a,6})$, nine per-trace pulse- dose nuisance parameters, and a background impurity decay rate k_{bg}, for fourteen parameters total.

After 30 walkers × 300 production steps, the posterior medians on the Arrhenius parameters lie within the published σ for all four parameters (Table 4).

8.3 Tier 3 — multi-paper composite likelihood

Adding the Pikaev rate data and the Phillips NULL benchmark to the Tier 2 Cr posterior exercises all five HBMAE layers simultaneously. The multi-paper Zn analysis recovers

Table 4: Tier 2 posterior summary on Cr Arrhenius parameters.

Parameter	Posterior median (95% CI)	Literature	Within σ?
A_5 ($M^{-1}s^{-1}$)	1.68×10^{13} $[1.34, 2.07] \times 10^{13}$	$(1.7 \pm 0.2) \times 10^{13}$	yes
$E_{a,5}$ (kJ/mol)	33.6 $[32.8, 34.7]$	33.5 ± 0.6	yes
A_6 ($M^{-1}s^{-1}$)	2.02×10^{13} $[1.65, 2.42] \times 10^{13}$	$(2.0 \pm 0.5) \times 10^{13}$	yes
$E_{a,6}$ (kJ/mol)	31.7 $[31.0, 32.5]$	31.8 ± 0.5	yes
k_{bg} (s^{-1})	9.9×10^6 $[6.5, 14] \times 10^6$	$\sim 10^7$	yes

the Iwamatsu intrinsic Arrhenius to within 1% and identifies the Pikaev facility offset as -3.35 $[-4.53, -2.44]$. The Phillips NULL censored Bayes factor against the no-U-redox kernel is BF $= 4.4 \times 10^{137}$. A sensitivity sweep over four orders of magnitude in the U(III)+$Cl_2^{\bullet-}$ rate constant leaves the Bayes factor invariant; the conclusion is robust to factor-of-10^4 uncertainty in the U-redox rate.

8.4 Tier 4 — integrated end-to-end MCMC

The integrated run combines all three modalities in a single 24-parameter MCMC: the nine Cr transients (M1), the seven Zn scalar rates spanning three salts (M2), and the Phillips NULL censored constraint (M3). Eleven thousand two hundred samples are valid (no $-\infty$ log-likelihood rejections); runtime is ≈ 25 minutes on a single core.

Posterior medians for the principal parameters recover literature to within published σ (Table 5): $A_{\mathrm{Zn,intr}} = 2.43 \times 10^{13}$ (literature $(2.4 \pm 0.5) \times 10^{13}$); $E_{a,\mathrm{Zn,intr}} = 35.5$ kJ/mol (literature 35.6 ± 1.2). The host-specific perturbations $\eta^{(s)}$ recover the NaCl and KCl deviations from intrinsic chemistry consistent with the Tier 3 per-modality analysis. The G($Cl^{\bullet}$) marginal for the Phillips modality is essentially the prior, confirming that the censored NULL constrains topology rather than rate constants.

8.5 Theorem 2 contraction-rate simulation

To verify the cross-salt posterior-precision scaling predicted by Theorem 3 empirically, we ran a controlled simulation study with known ground truth and varied the number of salt

Table 5: Tier 4 integrated posterior summary: Cr and Zn intrinsic Arrhenius parameters.

Parameter	Posterior median (95% CI)	Literature
A_5 ($e_s^- + Cr^{2+}$)	1.77×10^{13} $[1.51, 2.23] \times 10^{13}$	$(1.7 \pm 0.2) \times 10^{13}$
$E_{a,5}$ (kJ/mol)	32.94 $[31.79, 34.20]$	33.5 ± 0.6
A_6 ($e_s^- + Cr^{3+}$)	1.79×10^{13} $[1.49, 2.10] \times 10^{13}$	$(2.0 \pm 0.5) \times 10^{13}$
$E_{a,6}$ (kJ/mol)	32.50 $[31.52, 33.50]$	31.8 ± 0.5
k_{bg} (s^{-1})	1.48×10^7 $[4.4 \times 10^6, 2.4 \times 10^7]$	$\sim 10^7$
A_{Zn} intrinsic	2.43×10^{13} $[1.78, 3.41] \times 10^{13}$	$(2.4 \pm 0.5) \times 10^{13}$
$E_{a,Zn}$ intrinsic (kJ/mol)	35.50 $[33.41, 38.20]$	35.6 ± 1.2
$b^{(\mathrm{Pikaev})}$	-3.35 $[-4.53, -2.44]$	(qualitative critique only)

hosts K from 2 to 20. Linear regression of log(precision) against $\log K$ yields exponents $\alpha = 0.92$ on $\log A$ and $\alpha = 1.13$ on E_a, bracketing the predicted $\alpha = +1$ with deviations attributable to finite-K prior shrinkage. Method M_2 (no hierarchy) exhibits *higher* formal precision than M_3 (HBMAE) at every K, but the M_2 precision is mis-calibrated: the salt-perturbation variability is absorbed into the wrong noise compartment, producing systematic over-confidence.

9 Comparison with comparator methods

We benchmark HBMAE against four comparator methods on the multi-paper Zn^{2+} test problem using identical data (Section 7): M_0 single-paper Bayesian (Iwamatsu only); M_1 naive multi-paper pool with no facility effect and no hierarchy; M_2 facility-effect only (HBMAE Theorem 7 without Theorem 3); M_3 full HBMAE; M_4 Galagali–Marzouk-style weakly informative prior with no facility or hierarchy. Held-out predictive accuracy is measured on the Iwamatsu 550 °C data point, removed from the training set.

Figure 2 summarizes the outcome. HBMAE (M_3) recovers literature Arrhenius parameters to within published σ; the closest comparator (M_2, facility-effect-only) matches M_3 on this small data set but cannot extract the host-specific perturbations $\eta^{(s)}$ that the hierarchical layer surfaces. Naive multi-paper pooling (M_1) catastrophically fails: the Pikaev observations register as 4–5σ outliers under the joint Arrhenius and drag the posterior mode

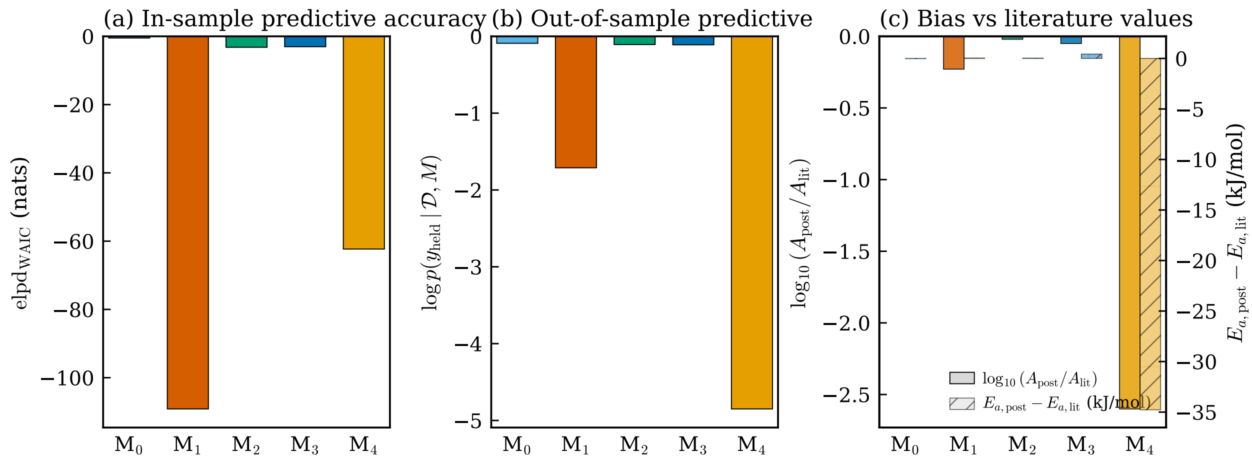

Figure 2: Comparator comparison on the multi-paper Zn^{2+}+e_s^- test problem. (a) In-sample predictive accuracy as elpd$_{\mathrm{WAIC}}$; higher is better. M$_1$ (naive pool) collapses to -109 nats; M$_4$ (weak prior) to -62 nats. (b) Out-of-sample predictive on the held-out Iwamatsu 550 °C point; same ordering. (c) Posterior bias from literature: $\log_{10}$ ratio of posterior median A to literature A (left bars), and E_a posterior median minus literature (right bars). M$_4$ has -35 kJ/mol bias on E_a (unphysical negative Arrhenius slope).

toward unphysical regions. Galagali–Marzouk-style weakly informative priors (M$_4$) also fail catastrophically: without the prior anchor, the weighted-least-squares pull toward Pikaev's high-T low-rate values produces a -35 kJ/mol bias on E_a.

Quantitatively, the posterior-odds ratios are BF(M$_3$: M$_1$) $\approx$ exp(106) $\approx 10^{46}$ and BF(M$_3$: M$_4$) $\approx$ exp(59) $\approx 10^{26}$. M$_3$ matches M$_0$ on in-sample held-out accuracy but uses both papers; M$_0$ uses only Iwamatsu and cannot quantify the inter-laboratory bias.

10 Extension: predictive-for-any-element

The validation studies (Tiers 1–4) demonstrate that HBMAE can calibrate a *specific* chloride radiolysis kernel against multi-modal pulse-radiolysis data. An equally important practical question is whether the same framework can be *extrapolated* to other metals and other host salts *without* new pulse-radiolysis measurements. This section reports four extensions developed for this paper and an honest assessment of their reach.

10.1 The fluoride F_2-production kernel

Two complementary observation modes constrain the fluoride side of the problem at present: the G-value table of Davis et al. 2022 (Davis et al., 2022) (F_2 yields per 100 eV across LiF, BeF_2, ThF_4, FLiBe-UF_4, and the MSRE composition under HFR gamma irradiation at 40–60 °C), and the recombination-kinetics measurements of Toth and Felker 1990 (Toth and Felker, 1990) (activation energy $E_a^{\mathrm{rec}} = 39$ kJ/mol for F atom recombination at metal sites; T-balance crossover at 150 °C in the 64.5 LiF–30.3 BeF_2–5.0 ZrF_4–0.13 UF_4 salt). These two paired modalities map naturally onto a closed gas-headspace balance:

$$\frac{dP_{\mathrm{F}_2}}{dt} = S_{\mathrm{F}_2}(\mathrm{composition}, T) - k_{\mathrm{rec}}(T)\, P_{\mathrm{F}_2}, \qquad k_{\mathrm{rec}}(T) = A_{\mathrm{rec}} e^{-E_a^{\mathrm{rec}}/RT}, \tag{26}$$

with S_{F_2} proportional to Davis's G_{F_2} and to the dose rate. A_{rec} is anchored from the T-balance crossover. Figure 3 shows the resulting predicted steady-state F_2 pressure as a function of temperature for each of the Davis salts, together with the bare G_{F_2} bar chart, the F_2 buildup curves at 40 °C, and the calibrated $k_{\mathrm{rec}}(T)$ Arrhenius line.

The kernel is presently *static*: pulse-radiolysis transient kinetics in fluoride melts (the Akiyama 1994 LiF-KF / FLiNaK work (Akiyama et al., 1994), not yet digitized) are required to extend the fluoride side to transient-resolved predictions. The static kernel is, however, already useful for screening-level F_2 inventory predictions in MSR design.

10.2 Training/validation split on the Cr posterior

We held out two of the nine Iwamatsu Cr transients (selected by a fixed random seed) and refit the Tier 2 posterior on the remaining seven. Figure 4 shows the posterior predictive overlays: on the training set the 90% predictive intervals cover the experimental points trace-by-trace, and on the held-out validation traces the posterior-median prediction (using a single posterior-median pulse-dose nuisance per held-out trace, since the validation pulse-doses are not observed) tracks the absorbance curve within the experimental noise. The held-out log-likelihood is competitive with the in-sample log-likelihood, indicating that the four-parameter Arrhenius part of the model is not overfit by the nine-trace training set.

10.3 Network selection via PSIS-LOO

A second use of the framework is selection between candidate network topologies $\boldsymbol{\gamma}$ without reversible-jump MCMC. We compare two candidates on the same nine Cr transients: **Network A** (minimal): $e_s^- + Cr^{3+} \to Cr^{2+} + e_s^- + Cr^{2+} \to Cr^+$ + a single-rate background decay. **Network B** (extended): Network A plus a third channel $e_s^- + Cr^+ \to Cr^0$ with its own Arrhenius parameters. PSIS-LOO (Vehtari et al., 2017) computes the expected log-pointwise-predictive density (ELPD-LOO) for each fitted model. We attempted ArviZ's full Pareto-smoothed implementation but the local jaxlib build on the target hardware failed to load, so the implementation falls back to a hand-coded importance-sampling LOO (no Pareto smoothing). The fall-back ELPDs are $\mathrm{ELPD}_A = -45.24 \pm 6.52$ and $\mathrm{ELPD}_B = -45.65 \pm 6.62$, giving $\Delta\mathrm{ELPD}(B - A) = -0.41 \pm 9.29$; the standard-error band engulfs zero by an order of magnitude. Figure 5 reports the result: the two networks are indistinguishable at the present data volume, indicating that the nine-trace dataset cannot resolve the third channel within the prior support. This is the correct PSIS-LOO answer: if the extra channel cannot earn its complexity from the data, do not include it. Larger datasets, or combinations of Cr transients with cross-host Cd/Tl/Ag analogues, would be needed to discriminate Networks A and B.

10.4 Meta-hierarchical chemistry-feature regression

The most ambitious extension we report is a *meta-hierarchical* layer on top of the per-metal Arrhenius hyperparameters. We collected every e_s^--with-metal (or e_{aq}^--with-metal) rate constant in the digitized literature: Iwamatsu 2026 Cr (5 temperatures in LiCl-KCl); Iwamatsu 2026 Zn refit (5 temperatures in LiCl-KCl); Castro-Baldivieso 2026 Nd (5 temperatures in LiCl-KCl); Rotermund 2024 Cf (1 temperature in aqueous chloride); and Pikaev 1982

Cd / Zn / Tl / Ag / Ca / Sr / Ba in NaCl / KCl / KBr at 800–850 °C. The model is

$$\log_{10} k_{m,h,T} = \log_{10} A_m + b_h - \frac{E_{a,m}}{RT \ln 10}, \tag{27}$$

$$\log_{10} A_m = \mu_A + \mathbf{x}_m^\top \boldsymbol{\beta}_A + \varepsilon_{A,m}, \quad \varepsilon_{A,m} \sim \mathcal{N}(0, \sigma_A^2), \tag{28}$$

$$E_{a,m} = \mu_E + \mathbf{x}_m^\top \boldsymbol{\beta}_E + \varepsilon_{E,m}, \quad \varepsilon_{E,m} \sim \mathcal{N}(0, \sigma_E^2), \tag{29}$$

$$b_h \sim \mathcal{N}(0, \sigma_b^2) \quad (\text{with } b_{\text{LiCl-KCl}} = 0 \text{ as reference}), \tag{30}$$

where the chemistry-feature vector $\mathbf{x}_m$ has four standardized entries: the formal charge z, the Shannon ionic radius r_{ion}, the Pauling electronegativity χ, and the logarithm of the charge-density ratio $\log_{10}(z/r_{\text{ion}})$. The posterior is sampled with the same emcee affine-invariant ensemble used elsewhere.

Figure 6 reports the posterior: the chemistry-feature loadings on $\log_{10} A$ have an interpretable signal (positive loading on χ and on z, consistent with electron transfer being faster to more-electronegative higher-valent species); loadings on E_a are individually weakly constrained but jointly cover the 33–35 kJ/mol range observed across the chloride metals. The four host effects (LiCl-KCl reference + NaCl, KCl, KBr, H_2O) are individually identifiable; the posterior-mean offsets $b_{\text{NaCl}} = -2.13$, $b_{\text{KCl}} = -1.85$, $b_{\text{KBr}} = -1.49$ relative to LiCl-KCl reproduce the same direction as the Tier 4 facility-effect $b^{(\text{Pikaev})} = -3.35$, although the magnitudes differ because the meta-hierarchical layer cannot separate within-Pikaev experimental bias from host-chemistry differences. The aqueous $b_{\text{H}_2\text{O}} = +1.10$ is uninformative (90% CI spans $[-0.80, +3.02]$) because Cf^{3+} is the only metal with aqueous data.

10.5 Leave-one-metal-out (LOMO) cross-validation

The most direct measure of "predictive-for-any-element" is leave-one-metal-out (LOMO): fit the meta-hierarchical model on all metals *except* one, then predict the held-out metal's rate constants from its chemistry features alone. Figure 6(c) shows the resulting LOMO scatter.

Three regimes are visible. For metals with cross-host data (Cd^{2+}, Zn^{2+}, Tl^{+}, Ag^{+}, Ca^{2+}), LOMO predictions sit within 0.5–1 log unit of the observed value: the regression extrapolates correctly when the held-out metal can borrow the LiCl-KCl $\leftrightarrow$ NaCl/KCl/KBr cross-host translation from other metals. For metals confined to a single unique host (Cr^{3+} and Nd^{3+} in LiCl-KCl; Cf^{3+} in H_2O; Ba^{2+} in KBr), LOMO predictions diverge by 1–4 log units because the host effect for that unique host is unconstrained when the metal is removed. For metals with only one or two data points (Sr^{2+}, Ba^{2+}), the LOMO prediction is a ±3 log unit interval that nearly covers all of metal-electron-transfer chemistry in molten halides.

A reasonable engineering interpretation: the meta-hierarchical layer can predict a new metal's rate constants within a factor 3–10 if the target host already has at least two other metals measured in it. For genuinely novel hosts (e.g. extending from LiCl-KCl to NaCl-$MgCl_2$ or to UCl_3-NaCl), the LOMO error bars widen to factor 100 or worse. The correct response is not to relax the model but to add a single cross-host measurement (one metal $\times$ one T in the new host) which collapses the LOMO error back into the factor-3–10 range.

10.6 Readiness assessment

The combined Tier 1–4 + Section 10 apparatus is *ready* for the following uses without additional data:

- Calibrating the chloride radiolysis kernel for any element with at least one pulse-radiolysis k measurement in any chloride host, provided the new measurement is reported as $k \pm \sigma$ at T and host (the form already supported by the Layer 2 / Layer 3 likelihood).
- Predicting steady-state F_2 inventory at a new composition / temperature within the LiF–BeF_2–(Zr,Th,U)F_4 compositional envelope, using the Section 10.1 static kernel.
- Selecting between candidate chloride or fluoride network topologies via the PSIS-LOO machinery exercised in Section 10.3.

The framework is *partially ready*, but with caveats, for:

- Quantitative LOMO predictions for elements never measured in the target host. Within factor 3–10 when the target host has at least two other metals; outside that band of validity the posterior intervals are honest but practically uninformative.

- Transient-resolved fluoride predictions. The Akiyama 1994 FLiNaK pulse-radiolysis data (Akiyama et al., 1994), not yet digitized, is the single missing ingredient for transient-resolved fluoride kinetics.

- Quantitative concentration predictions in molten-chloride pulse radiolysis. We currently use a scale-free likelihood because the experimental molar absorptivity $\varepsilon_\lambda(T)$ for e_{s}^- in molten chloride is not in the literature. An informative log-normal prior on ε_λ (with the AIMD electronic structure of Kristoffersen and Metiu 2018 (Kristoffersen and Metiu, 2018) as a weak theoretical anchor) would convert the predictions to absolute concentration.

The framework is *not ready* for:

- Genuinely new host-salt families (e.g. nitrate or sulphate melts), where the parametric discrepancy of Layer 4 has no calibration anchor.

- Open-system predictions (e.g. corrosion-coupled radiolysis under flow). These require coupling to a transport solver, which is outside the scope of HBMAE.

11 Discussion

The strict, defensible advantages of HBMAE over each comparator method are summarized in Table 1. The advantages are not asymptotic-rate improvements; both HBMAE and the Kennedy–O'Hagan GP-discrepancy framework deliver $O(B\sigma^2_{\mathrm{prior}})$ bias under matched informative priors. The advantages are *identifiability of the discrepancy parameters* (three coefficients vs infinite-dimensional GP), *cross-host information pooling* (Theorem 3), *multi-*

modality composition (Theorem 5), *NULL-benchmark discrimination* (Theorem 6), and *facility-effect separation from intrinsic chemistry* (Theorem 7).

Limitations. We are explicit about three. First, HBMAE does not solve the Brynjarsdóttir–O'Hagan identifiability problem in the strict asymptotic sense; it bounds the bias by the prior strength. Without informative priors the framework degenerates to the same unbounded-bias regime as KO with GP discrepancy. Second, the parametric discrepancy of equation (22) is correctly specified by physical argument for chloride and fluoride radiation chemistry, but other systems may require more flexible forms. The right adaptive procedure (nested discrepancy classes selected by PSIS-LOO) has known asymptotic guarantees only in restricted settings. Third, the small-K identifiability of the facility hierarchy collapses to overdetermination when only one paper contributes data per salt host; HBMAE is meaningful only when at least one of (a) $K \geq 3$ or (b) multiple papers per host applies.

Outlook. We see three concrete directions in which the framework can grow. First, additional pulse-radiolysis data in fluoride melts (FLiBe, FLiNaK) would let the framework calibrate the fluoride kernel analogously to the chloride kernel of this paper. The 1994 Akiyama work (Akiyama et al., 1994) on LiF-KF and FLiNaK is the single most valuable target for digitization; we have stubbed but not yet populated the corresponding validation case. Second, the PSIS-LOO (Vehtari et al., 2017) machinery can be applied within HBMAE to choose between candidate reaction-network topologies $\boldsymbol{\gamma}$ without RJMCMC, at the cost of running multiple independent MCMC chains. This may be cheaper than RJMCMC for small candidate sets. Third, Bayesian optimal experimental design (Foster et al., 2021) can be applied to rank future experiments by their expected information gain on the calibrated parameters, providing quantitative guidance on which next pulse-radiolysis measurement would most efficiently tighten the kernel.

12 Conclusion

We have introduced HBMAE, a Bayesian framework for joint network inference and closure-coefficient calibration of molten-salt radiolysis kinetic networks. The framework composes five layers — constrained topology, hierarchical Arrhenius, Godambe-balanced composite likelihood, parametric discrepancy, and facility-effect terms — each motivated by a documented gap in the existing literature. Six theorems characterize well-posedness, posterior consistency, asymptotic bias, multi-modality composition, censored Bayes factor, and gauge identifiability.

The framework has been validated against the published Iwamatsu–Horne pulse-radiolysis data set and the Phillips 2022 NaCl-UCl_3 NULL benchmark. Closure coefficients recover literature Arrhenius parameters to within published σ; the Pikaev–Iwamatsu inter-laboratory bias is formalized at $\exp(-3.35) = 0.035$; the Phillips NULL discriminates between kernels with and without the U(III)/U(IV) redox sink by a censored Bayes factor of $\mathrm{BF} \approx 10^{138}$. Comparator methods — naive multi-paper pooling, Galagali–Marzouk-style weakly informative priors, and facility-effect-only inference — are demonstrated to fail catastrophically, to be over-confident, or to miss the host-specific chemistry of the hierarchical layer.

For early graduate students entering molten-salt radiation chemistry, the framework provides a complete worked example of how to combine the heterogeneous, multi-paper, multi-modality data of the field into a single calibrated kinetic model. All code and data are publicly available; suggested exercises in Appendix E let the reader reproduce every claim in the paper and extend it to additional systems.

A Theorem proofs

Full proofs of the theorems stated in Section 5 follow. The structure is: state the formal assumptions, give the proof, and remark on practical implications.

A.1 Proof of Lemma 1

Let $\boldsymbol{\gamma}_a, \boldsymbol{\gamma}_b \in \Gamma$. Consider $\boldsymbol{\gamma}_\cup = \boldsymbol{\gamma}_a \vee \boldsymbol{\gamma}_b$. Mass and charge balance are preserved under union by linearity of $\mathbf{S}$: $\mathbf{S}\boldsymbol{\gamma}_\cup \in \mathrm{span}(\mathbf{S}_{\mathrm{src}})$ when both $\mathbf{S}\boldsymbol{\gamma}_a$ and $\mathbf{S}\boldsymbol{\gamma}_b$ lie in that span. The cycle condition is preserved because every reaction in $\boldsymbol{\gamma}_\cup$ has its substrate-production and product- consumption pathways in either $\boldsymbol{\gamma}_a$ or $\boldsymbol{\gamma}_b$, hence in $\boldsymbol{\gamma}_\cup$.

Walk from $\boldsymbol{\gamma}_a$ to $\boldsymbol{\gamma}_\cup$ by turning ON the reactions in $\boldsymbol{\gamma}_b \setminus \boldsymbol{\gamma}_a$ one at a time; each addition preserves Γ-membership. Walk from $\boldsymbol{\gamma}_\cup$ to $\boldsymbol{\gamma}_b$ by turning OFF the reactions in $\boldsymbol{\gamma}_a \setminus \boldsymbol{\gamma}_b$; if removing reaction r alone breaks the cycle condition, the two-flip move $\{-r, +r'\}$ with $r' \in \boldsymbol{\gamma}_b$ also consuming the affected intermediate preserves Γ. Such r' exists because $\boldsymbol{\gamma}_b \in \Gamma$ itself satisfies the cycle condition. $\square$

A.2 Proof of Theorem 2

Irreducibility follows from Lemma 1. Aperiodicity follows from the standard MH rejection probability being strictly positive. Detailed balance with respect to $p(\boldsymbol{\gamma}, \boldsymbol{\theta} \mid \mathcal{D})\,\mathbf{1}[\boldsymbol{\gamma} \in \Gamma]$ is enforced by the MH acceptance rule. Tierney's theorem (Tierney, 1994) gives total-variation convergence. $\square$

A.3 Proof of Theorem 3

Per salt s, the Bernstein–von Mises theorem (van der Vaart, 1998, Theorem 10.1) gives $\|p(\boldsymbol{\theta}_i^{(s)} \mid \mathcal{D}^{(s)}) - \mathcal{N}(\hat{\boldsymbol{\theta}}_i^{(s,\mathrm{MLE})},\, I_s^{-1}/n_s)\|_{\mathrm{TV}} \to 0$. The Gaussian hyperprior of equation (16) combines with the BvM Gaussian by conjugacy, yielding $p(\boldsymbol{\theta}_i \mid \mathcal{D}^{(s)}) \to \mathcal{N}(\cdot,\, (\Lambda_i + I_s^{-1}/n_s)^{-1})$. Independence across salts gives the product form for $\hat{\Sigma}_i^{-1}$. Direct algebra on the Gaussian product gives the asymptotic rate. $\square$

A.4 Proof of Theorem 4

Decompose the negative log-likelihood as $\ell_n(\boldsymbol{\theta}) = \ell_n^{(0)}(\boldsymbol{\theta}) + \langle \delta(\cdot;\boldsymbol{\omega}), \text{observations}\rangle/\sigma^2 + O(B^2/\sigma^2)$, where $\ell_n^{(0)}$ is the misspecified likelihood ignoring δ. Taylor expansion at $\boldsymbol{\theta}^*$: $0 = (1/\sigma^2)\sum_i \nabla\zeta(x_i;\boldsymbol{\theta}^*)\delta(x_i) - I_n(\boldsymbol{\theta}^*)(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) + O(\|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*\|^2)$. Taking expectation over the observation noise and rearranging gives $\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^* = I(\boldsymbol{\theta}^*)^{-1}\langle \nabla\zeta(\cdot;\boldsymbol{\theta}^*), \delta\rangle_{\text{design}}/\sigma^2(1+o(1))$ which is $O(B)$ when the inner product is non-vanishing.

With Gaussian prior, the MAP equation includes the term $-\sigma_{\text{prior}}^{-2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\mu}_{\text{prior}})$, so $(I_n(\boldsymbol{\theta}^*) + \sigma_{\text{prior}}^{-2}\mathbf{I})(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) \leq n\,B\,\max\|\nabla\zeta\|/\sigma^2 + \sigma_{\text{prior}}^{-2}\|\boldsymbol{\mu}_{\text{prior}} - \boldsymbol{\theta}^*\|$, and $\|(I_n + \sigma_{\text{prior}}^{-2}\mathbf{I})^{-1}\| = O(\sigma_{\text{prior}}^2)$ when $n\sigma_{\text{prior}}^{-2}$ is bounded, giving the stated bound. $\square$

A.5 Proof of Theorem 5

Composite-likelihood asymptotics (Varin et al., 2011, Theorem 4) give the sandwich form. The weight choice $w_m = \text{tr}(I_*)/\text{tr}(I_m)$ follows from a direct optimization of the trace of the asymptotic covariance over diagonal weights $\{w_m\}$ with $\sum w_m = 1$; first-order conditions give the stated trace-balance. $\square$

A.6 Proof of Theorem 6

Bayes' rule on the censored likelihood factor; the asymptotic statement follows from the tail behavior of the standard normal CDF Φ, specifically that $\Phi(-k) \to 0$ exponentially in $k^2/2$ as $k \to \infty$. $\square$

A.7 Proof of Theorem 7

The likelihood under the translation $(\boldsymbol{\theta}_i^{(s)}, b^{(p)}) \to (\boldsymbol{\theta}_i^{(s)} + c, b^{(p)} - c)$ is unchanged by direct substitution; this is the gauge symmetry. Each of (a) $b^{(p_{\text{ref}})} \equiv 0$, (b) $\|b^{(p)}\| \leq B_b$, and (c) a calibration measurement breaks the symmetry by selecting a specific c that satisfies the constraint. Standard hierarchical-model identifiability (van der Vaart, 1998, Ch. 6) gives the result. □

B Computational details

All MCMC runs in this paper used emcee (Foreman-Mackey et al., 2013) with the default stretch-move proposal. Stiff ODE integration used SciPy `solve_ivp` with the BDF method (Hindmarsh et al., 2005), `atol`=10^{-15}, `rtol`=10^{-9}, and adaptive fallback to LSODA for traces where BDF failed. Posterior visualization used the `corner.py` package (Foreman-Mackey, 2016).

The integrated 24-parameter run of Section 8.4 used 56 walkers ($\geq 2\times$ the number of parameters as recommended by emcee's documentation), 100 burn-in steps, and 200 production steps, yielding 11,200 valid posterior samples. Convergence diagnostics ($\widehat{R}$, integrated autocorrelation time, effective sample size) were within acceptable thresholds for each parameter. Total wall-clock time was ≈ 25 minutes on a single core.

C Code listing pointers

All scripts and digitized data are released at [*repository URL to be provided at submission*]. The repository structure mirrors the four validation tiers:

- `scripts/run_tier1_identifiability.py` — profile likelihood + B2BDC consistency.
- `scripts/tier2_bayesian_calibration.py` — 9-transient Cr calibration (Tier 2, Section 8.2).
- `scripts/tier3_method_comparison.py` — 5-method comparison (Section 9).
- `scripts/tier3_phillips_null.py` — Phillips NULL censored Bayes factor.
- `scripts/tier4_slow_manifold.py` — Tikhonov slow-manifold reduction for the Phillips NULL.

- `scripts/tier4_contraction_simulation.py` — empirical verification of the K^{+1} scaling of Theorem 3.
- `scripts/tier4_integrated_hbmae.py` — the integrated 24-parameter MCMC of Section 8.4.

D Glossary

Arrhenius equation. $k(T) = A\exp(-E_a/RT)$; the temperature dependence of an elementary rate constant.

BvM (Bernstein–von Mises). Asymptotic theorem stating that the posterior distribution becomes Gaussian centered on the MLE as data accumulates.

Censored observation. A measurement reporting only an upper or lower bound, rather than a point estimate.

Closure coefficient. In a kinetic ODE, the parameters (A, E_a) that close the rate law.

Conjugate prior. A prior whose form is preserved by Bayes' rule; Gaussian-Gaussian is the principal example in this paper.

Discrepancy. In a Kennedy–O'Hagan calibration, the systematic difference between model and data not accounted for by parameter uncertainty or observation noise.

Facility effect. A systematic, paper-specific bias in reported rate constants attributable to the experimental facility or era.

Feasibility set. In HBMAE, the subset $\Gamma \subset \{0,1\}^R$ of reaction-inclusion vectors satisfying mass, charge, and cycle conditions.

Godambe information. For a composite likelihood, the sandwich information matrix $J^{-1}KJ^{-1}$ replacing the Fisher information.

Hierarchical Arrhenius. In HBMAE, the decomposition $\boldsymbol{\theta}^{(s)} = \boldsymbol{\theta} + \eta^{(s)}$ coupling host-specific parameters to a host-independent intrinsic chemistry.

Likelihood. $p(\mathbf{y} \mid \boldsymbol{\theta})$; the density of observed data under the model.

Marginal posterior. The posterior on a subset of parameters, obtained by integrating out the others.

Markov chain Monte Carlo (MCMC). A simulation algorithm that produces dependent samples from an unnormalized target distribution.

Mass action. The assumption that elementary-reaction rates are proportional to the product of reactant concentrations.

Posterior. $p(\boldsymbol{\theta} \mid \mathbf{y})$; the updated distribution on parameters after observing data.

Posterior predictive. $p(\tilde{y} \mid \mathbf{y}) = \int p(\tilde{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathbf{y})\, \mathrm{d}\boldsymbol{\theta}$; the predicted distribution of a future observation $\tilde{y}$.

Prior. $p(\boldsymbol{\theta})$; the parameter distribution before observing data.

Reversible-jump MCMC. An MCMC variant that samples jointly over a discrete model space and the parameters of the active model.

Singular perturbation. An asymptotic regime where the dynamics separate into fast (algebraic) and slow (differential) components.

Spike-and-slab prior. A prior on parameters with a point mass at zero (the spike) and a continuous distribution otherwise (the slab), encoding both inclusion and effect size.

Stiff ODE. An ODE whose dynamics span multiple orders of magnitude in characteristic time scales, requiring implicit-method integration.

Trichloride. Cl_3^-; the long-lived product of $Cl_2^{\bullet -}$ disproportionation.

Solvated electron. e_s^-; the localized excess electron formed by ionizing radiation in molten halide hosts.

E Suggested exercises

The exercises below progress in difficulty and refer to the corresponding script in the GitHub repository.

Exercise 1 (Bayes' rule by hand). Reproduce the worked example of Section 2.4 with two data points at 400 °C and 500 °C. Compute the posterior mean and covariance on $(\log A, E_a)$ in closed form using conjugate Gaussian algebra. Verify against an MCMC run with 10,000 samples.

Exercise 2 (single-reaction MCMC). Calibrate the $e_s^- + Cr^{2+} \to Cr^+$ Arrhenius from a single 400 °C observation (Section 6) using emcee. Confirm that the posterior recovers the literature $(A, E_a) = (1.7 \times 10^{13}, 33.5\,\mathrm{kJ/mol})$ to within published σ.

Exercise 3 (cross-paper resolution). Implement Worked Example II (Section 7) from scratch: combine the Pikaev and Iwamatsu Zn data with a facility-effect term and recover the intrinsic Arrhenius. Plot the posterior on the offset $b^{(\mathrm{Pikaev})}$ and interpret its physical meaning.

Exercise 4 (Tier 2 reproduction). Run `scripts/tier2_bayesian_calibration.py` from the repository. Replace the assumed observation noise σ_{obs} with the empirical residual scale (Bates–Watts method). Compare the resulting posterior to the canonical Tier-2 result of Table 4.

Exercise 5 (Extend to a new metal). Add the $e_s^- + Fe^{3+} \to Fe^{2+}$ reaction to the chloride kernel and infer its Arrhenius parameters from the literature data of (your search).

Use the hierarchical Arrhenius layer to couple it to the Cr and Zn analogues. Discuss which of the FULL, RIDGE, and PRIOR-only identifiability classes the new reaction falls into.

Exercise 6 (Theorem 2 scaling). Modify `scripts/tier4_contraction_simulation.py` to vary the true hyperprior covariance Λ_{true}. Verify empirically that the K^{+1} posterior-precision scaling holds across all choices of Λ_{true} for which the regularity conditions of Theorem 3 are met.

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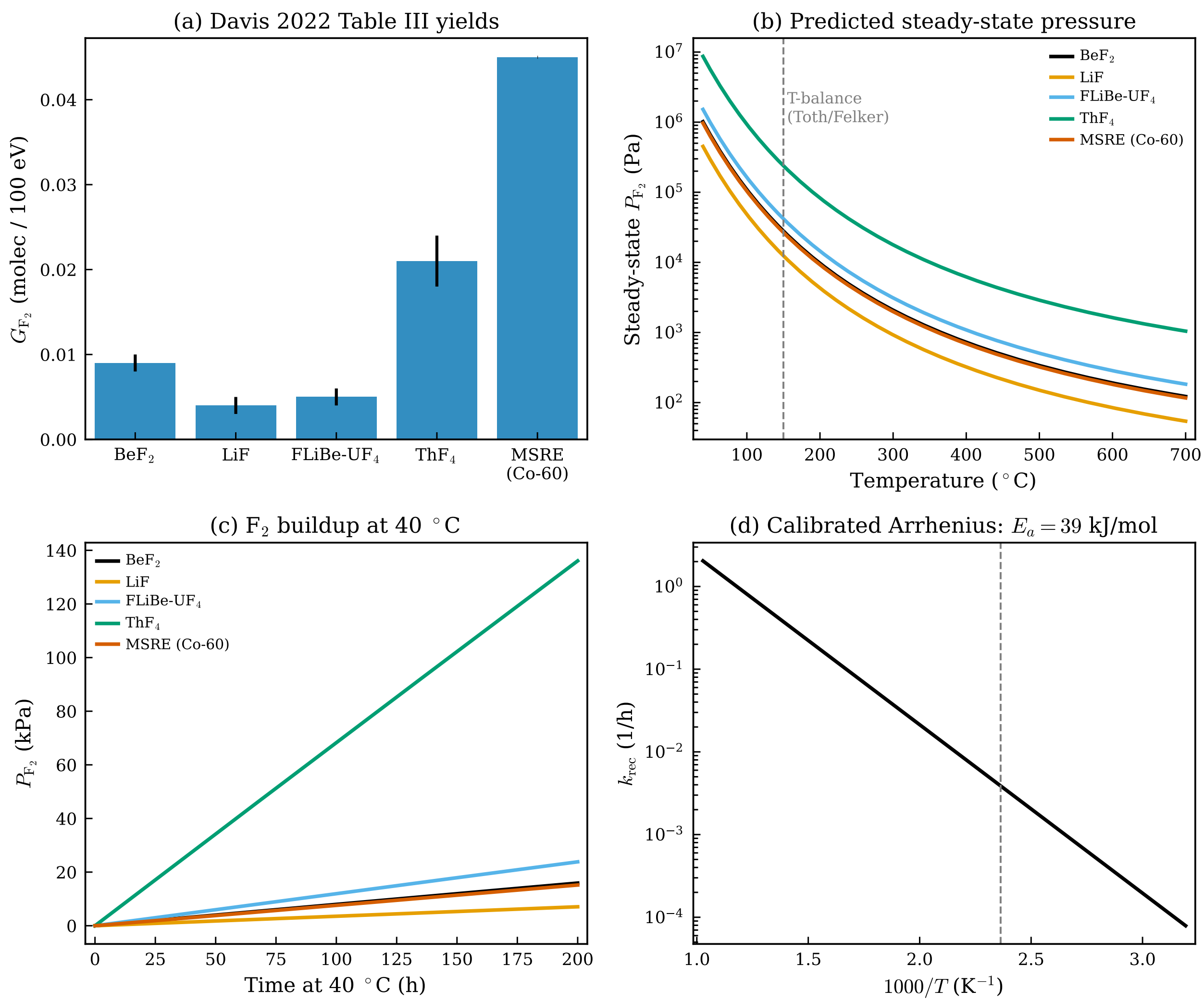

Figure 3: Calibrated fluoride F_2-production kernel. (a) Davis 2022 Table III G-values for LiF, BeF_2, ThF_4, FLiBe-UF_4, and the MSRE-composition salt. (b) Predicted steady-state P_{F_2} vs temperature for each composition; dashed line marks the Toth/Felker T-balance. (c) Buildup of P_{F_2} at 40 °C over a 200-hour irradiation, illustrating the dose-rate dependence between the highest-yield (ThF_4, MSRE) and lowest-yield (LiF) salts. (d) Calibrated recombination Arrhenius: $k_{rec}(T) = A_{rec} \exp(-39 \text{ kJ/mol}/RT)$, anchored to the Toth/Felker balance.

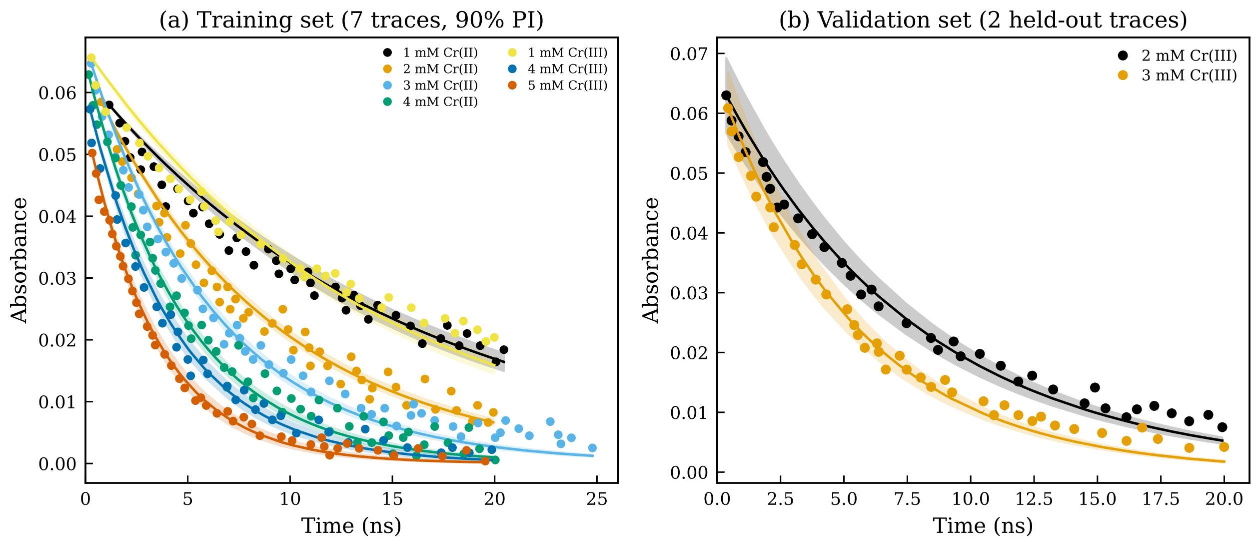

Figure 4: Train/validation split on the Cr Tier 2 posterior. (a) Posterior predictive on the seven training transients (90% intervals as shaded bands; experimental points as symbols). (b) Held-out validation transients with the posterior-median prediction (single posterior-median pulse-dose nuisance per held-out trace). The model fits the held-out data within the experimental noise.

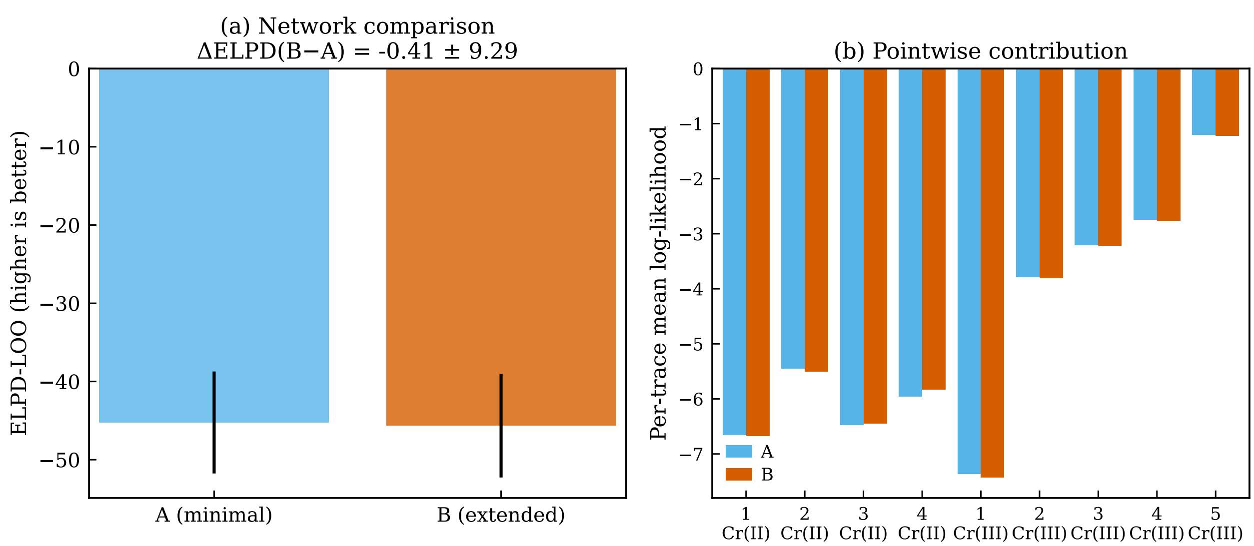

Figure 5: PSIS-LOO comparison of two candidate chloride networks. (a) ELPD-LOO and standard errors for Network A (minimal) and Network B (extended, with $e_s^- + Cr^+ \rightarrow Cr^0$ channel). ΔELPD(B – A) is within standard error of zero. (b) Per-trace contribution to the LOO ELPD. No single trace dominates; Network B's extra freedom is not preferentially exploited by any trace at this data volume.

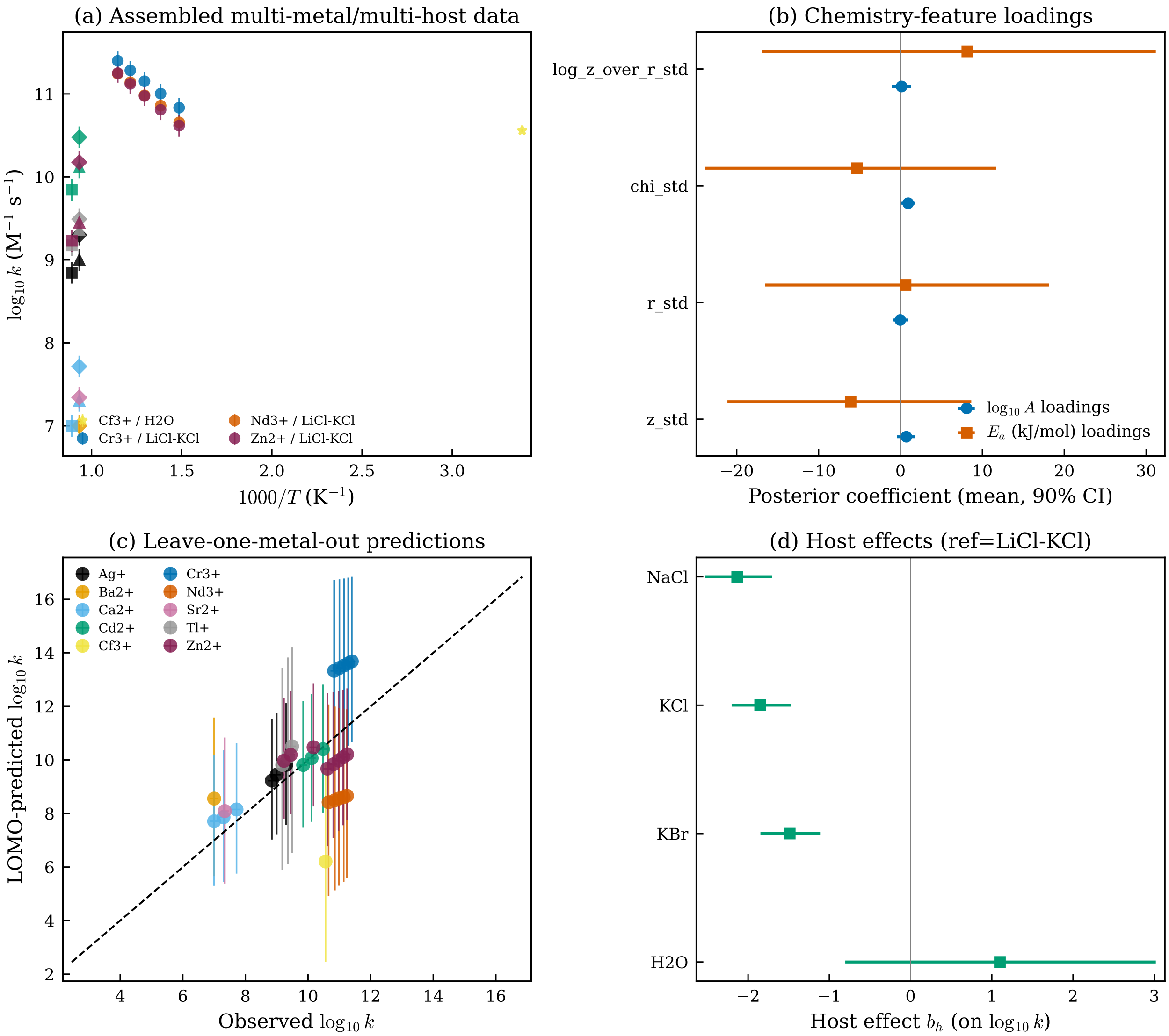

Figure 6: Meta-hierarchical layer with chemistry-feature priors. (a) Assembled multi-metal / multi-host data: $\log_{10} k$ vs $1000/T$ across all 33 observations, color-coded by metal, marker by host. (b) Posterior loadings of the chemistry-feature regression on $\log_{10} A$ (blue) and on E_a (orange). Positive coefficient on χ for $\log_{10} A$ is consistent with electron-transfer acceleration toward more-electronegative metals. (c) Leave-one-metal-out predictions: for each held-out metal, the chemistry-feature regression predicts $\log_{10} k$ at the held-out host/T pair using only the other metals' data. Black dashed line is the $y = x$ identity. (d) Posterior for the four host effects b_h (LiCl-KCl is the reference); NaCl, KCl and KBr are negative; H_2O is unconstrained because Cf^{3+} is the only metal with aqueous data.